UNIVERSITÄT BONN

Chapter 6:
Statistical Testing

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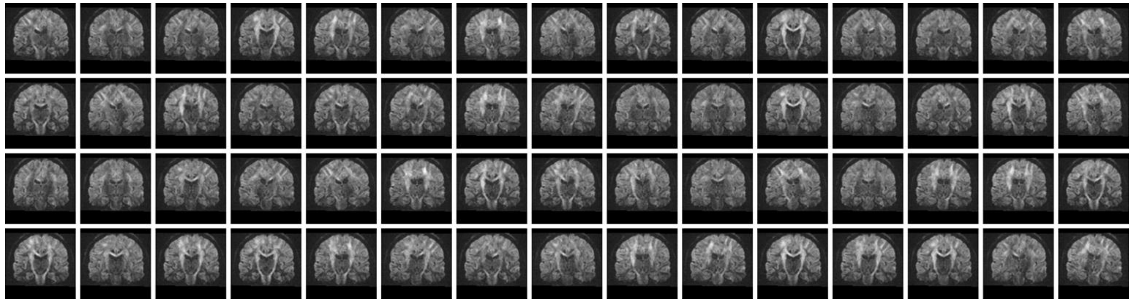
Bonn-Aachen
International Center for
Information Technology

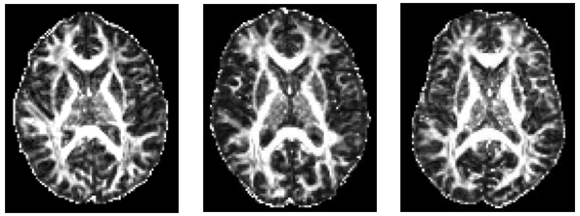
Image Acquisition and Analysis in Neuroscience

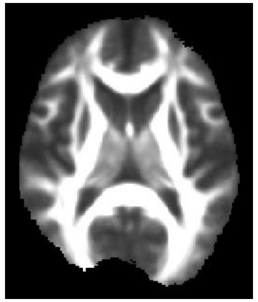
Acquisition

Reconstruction & Artifact Correction



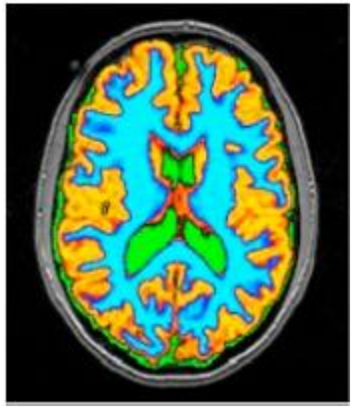
Registration



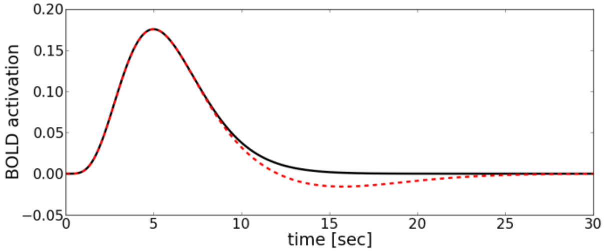


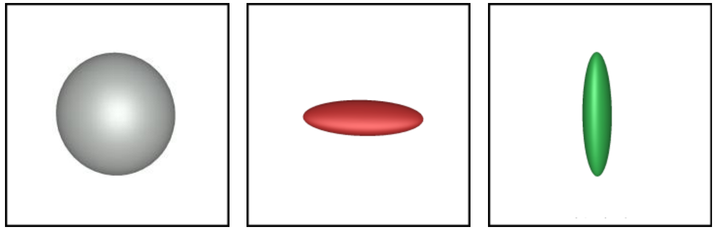
Atlas Construction

Segmentation

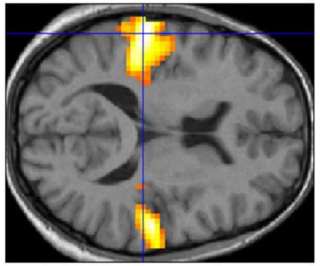


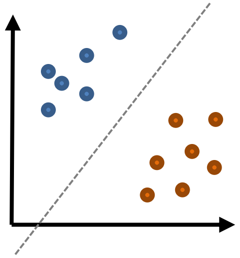
Mathematical Modeling





Hypothesis Testing





Machine Learning

6.1 Statistical Hypothesis Tests

Goal of Hypothesis Tests

- Now that we know about image acquisition, MR reconstruction, and image analysis (registration, segmentation), we would like to **draw scientific conclusions** from our data
  - Answer questions such as:
    - Can we observe changes in the thickness of the cortex associated with learning how to juggle?
    - Which brain regions are activated when counting?

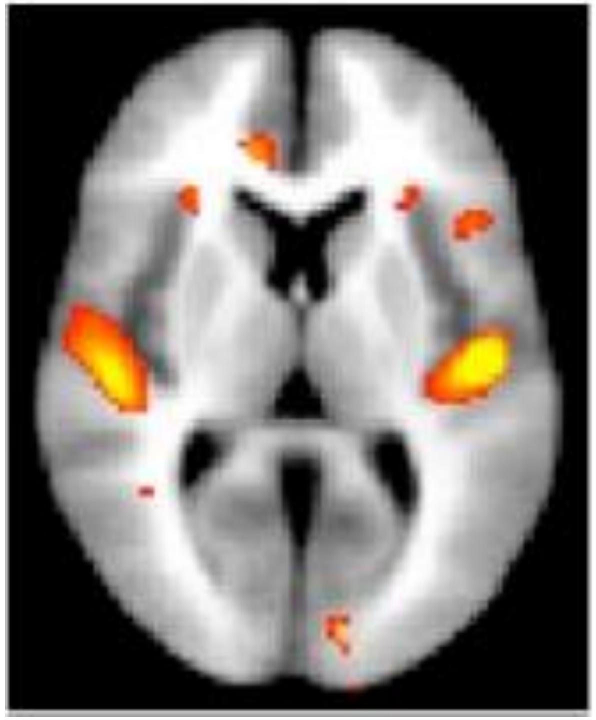
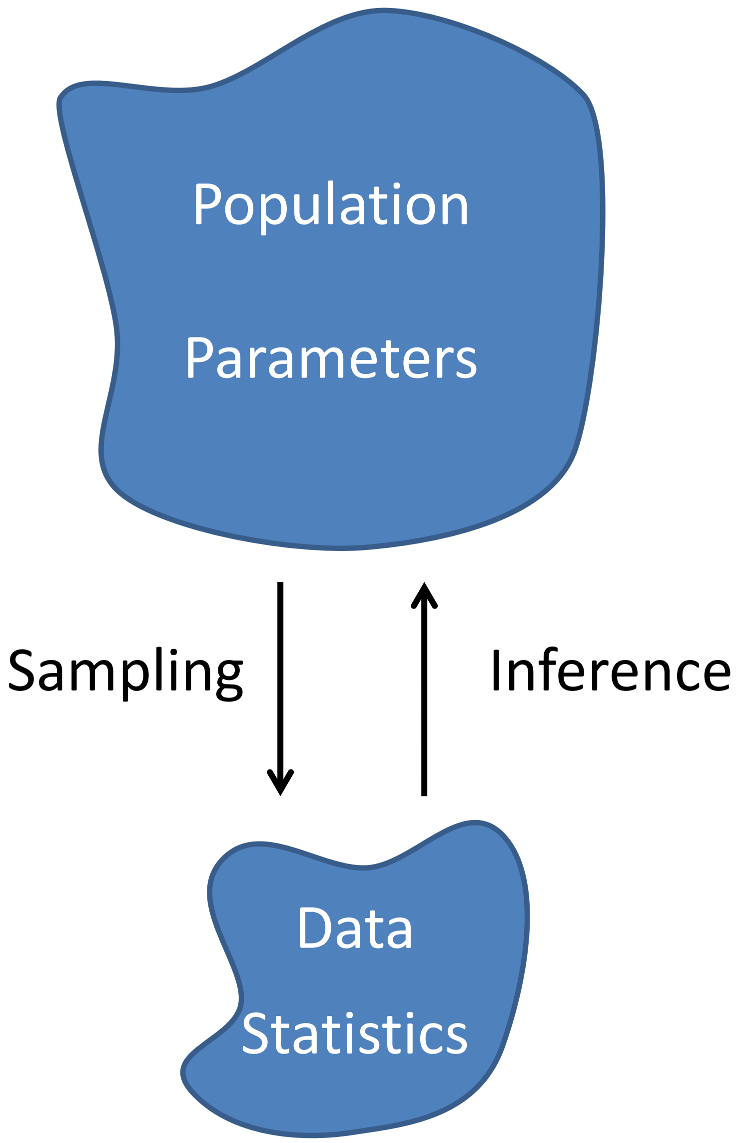


Image from Oxford brain fMRI lab

Population vs. Sample

- **Population:** Set about which we would like to make a statement
  - Quantity of interest is (assumed to be) governed by some statistical distribution (e.g., Gaussian) with certain parameters
  - *Example*: Mean gray matter volume in patients with Alzheimer's Disease compared to healthy controls
- **Sample:** Set of individuals for which we have observations
  - Can compute statistics on measured data
  - Would like to make statements about the parameters *of the partially observed population*
  - Would like to control the risk of making incorrect statements



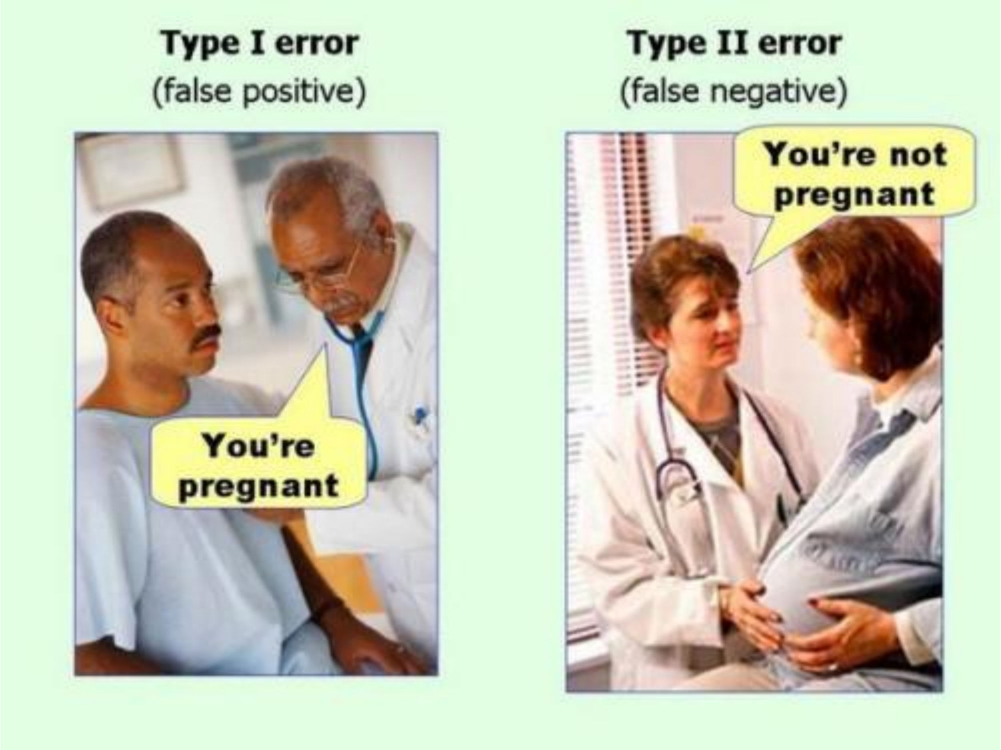
Hypothesis Testing: Basic Concept

- **Basic principle:** Reject a statement about the population parameters by showing that the observed data is highly unlikely if it holds
- **Null Hypothesis (H_0):** Statement that we are trying to reject
  - The opposite of our true research hypothesis
    - **Alternative hypothesis** H_A: "H_0 is false"
  - Often states that some factor *does not have an effect* on the data
    - e.g., "The changes from learning how to juggle are not visible in MRI."

Type I / Type II Errors

- **Type I error** (α): ***Reject*** a ***true*** null hypothesis
  - In science: reporting a false finding
  - Controlled by design of statistical test
  - Widely accepted level: α=5%

H_0	True	False
Reject	Type I	Correct
~~Reject~~	Correct	Type II

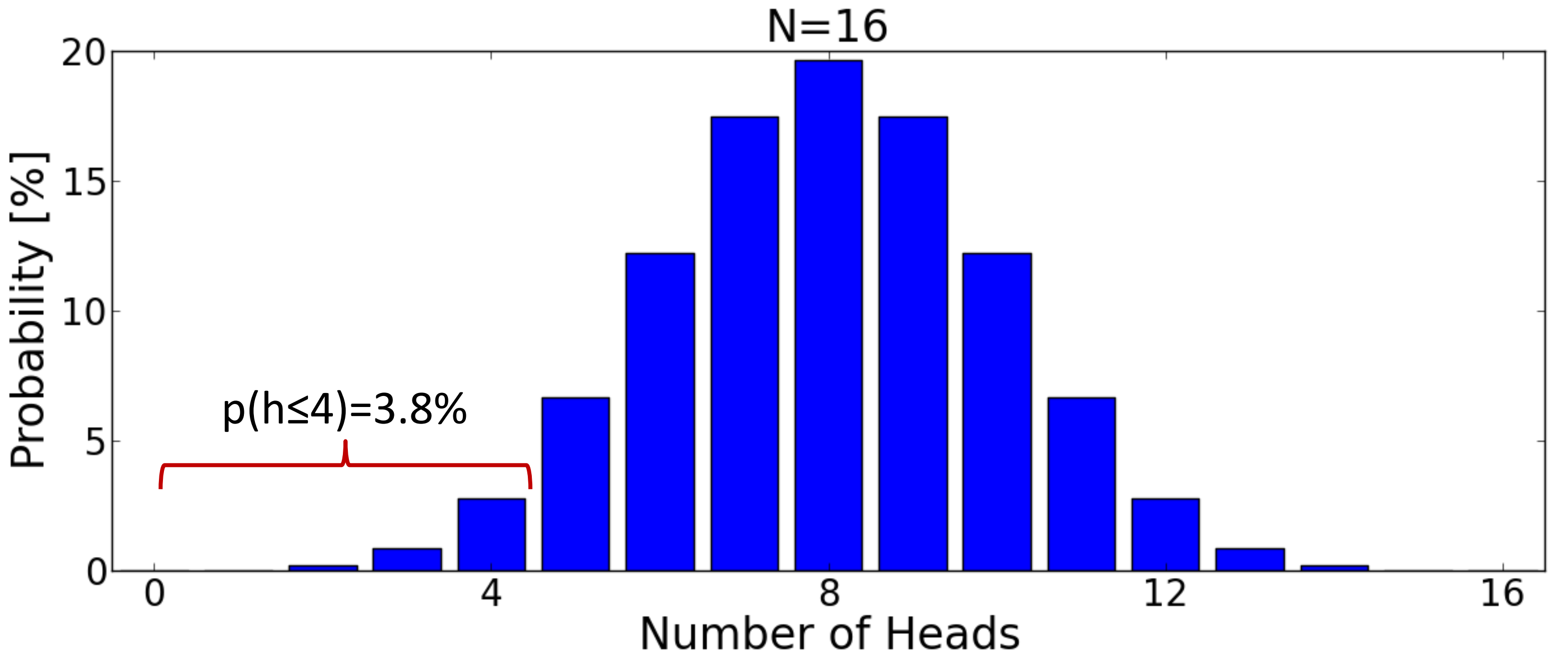


Source: https://effectsizefaq.com/ by Paul Ellis

- **Type II error** (β): ***Fail to reject*** a ***false*** null hypothesis
  - In science: Not being able to report a finding even though the alternate (research) hypothesis was correct, "insufficient power"

Example: Coin Toss

- **Null hypothesis:** A given coin is fair (equal probability of coming up heads or tails)
- **Evidence:** 4/16 tosses (25%) came up heads
- **Question:** Can we reject the null hypothesis?
- **Approach:** Under the null hypothesis, the number of heads follows a binomial distribution:



Definition of *p* value

- **Definition:** The **test statistic** is a statistical quantity that is derived from the sampled data and used as a basis of the test
  - *In our example:* Number of heads
- **Definition:** The ***p* value** is the *conditional* probability of observing the computed value of the test statistic or a more extreme value *if H_0 is true*
  - Often needs to be computed using software or using pre-computed lookup tables
  - Smaller *p*: Stronger evidence against H_0
  - Obtain a test with type I error α if we reject H_0 if $p<\alpha$
    - In neuroscience, $p<0.05$ is usually taken to be "significant"

One-sided vs. Two-sided Tests

- **One-sided *p:*** Sum of values (discrete test statistic) or area under the curve (continuous case) from the computed value to the end of the distribution
  - *In our example: p*=0.038
  - Suitable if original hypothesis was one-sided (e.g., we had a reason to assume that bias is in favor of tails)
- **Two-sided *p:*** Two times the one-sided *p*
  - Reason: Consider extreme values *at both tails*
  - *In our example: p*=2x0.038=0.076
  - Suitable if there was no prior evidence for a bias in any particular direction
  - *In our example,* p=0.076 is not significant

Caveat: Interpretation of *p* value

- The *p* value is
  - **Not** the probability α of committing a type I error
    - Within the framework of type I and type II errors, we have to fix α before the test and should only consider whether $p<\alpha$ (binary decision)
    - It is still informative to report the *p* value
  - **Not** a direct measure of effect size
    - *p* values also depend on the number of samples and the precision of estimates
  - **Not** the probability that the null hypothesis is true

Illustration: Screening for Rare Disease

- **Example:** You are tested for a rare, but fatal disease (affecting 1 person in 10.000) with a type I error of only 1%. The test rejects the null hypothesis that you are healthy. If you are healthy, the chance of this happening is 1 in 100! How much should you worry?
- **Answer:** Use Bayes' rule!
  Assume type II error is 5%.
  Remember that type I/II errors are **conditioned** on the (unknown) truth:

H_0	True	False
Reject	1%	95%
~~Reject~~	99%	5%

$\sum = 100\%$ $\sum = 100\%$

Illustration: Screening for Rare Disease

- To obtain **joint probabilities,** multiply first/second column by **prior probability** of being healthy/sick (99.99%/0.01%)

H_0	True	False
Reject	0.9999%	0.0095%
~~Reject~~	98.9901%	0.0005%

- To find the **posterior probability** of being sick, condition on the fact that the test turned out positive:

Probability of being healthy given the test result

H_0	True	False	
Reject	99.06%	0.94%	$\sum = 100\%$
~~Reject~~	99.9995%	0.0005%	$\sum = 100\%$

Hypothesis Tests: Caveats

- Hypothesis testing **ignores prior probabilities**
- Might reject the null **“for the wrong reason”**
  - Groups might differ in aspects other than the one we were interested in when designing the experiment
  - (Partial) remedy: Control for age, gender, handedness, years of education, disease
- Rejecting the null hypothesis does not imply an **unambiguous causal dependence**
- **Publication bias:** Often, H_0 is tested by several groups independently; typically, only cases in which it is rejected are published, inflating the type I error
  - Example of the problem of multiple comparisons

ASA's Statement on p-Values (Part 1)

Six statements published in 2016 by the American Statistical Association:

1. *p*-values can indicate how incompatible the data are with a specified statistical model.
2. *p*-values do not measure the probability that the studied hypothesis is true, or the probability that the data were produced by random chance alone.
3. Scientific conclusions and business or policy decisions should not be based only on whether a *p*-value passes a specific threshold.

ASA’s Statement on p-Values (Part 2)

4. Proper inference requires full reporting and transparency.
5. A *p*-value, or statistical significance, does not measure the size of an effect or the importance of a result.
6. By itself, a *p*-value does not provide a good measure of evidence regarding a model or hypothesis.

Reference: Wasserstein/Lazar, The American Statistician 70(2), 2016

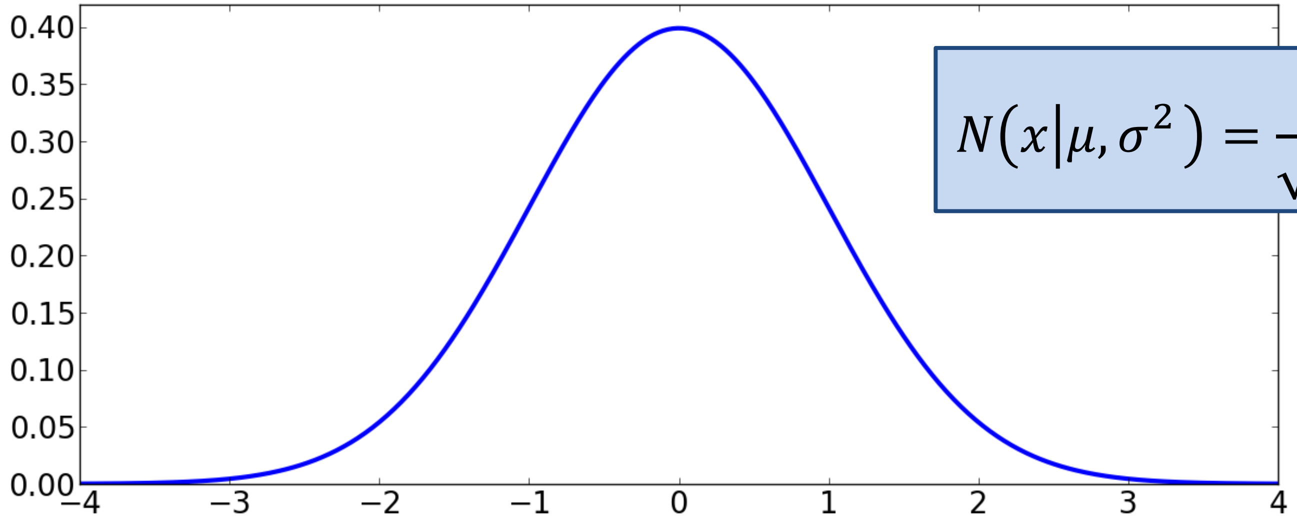
Summary: Hypothesis Testing

- Hypothesis testing rejects a null hypothesis H_0 based on the fact that the observed data is unlikely given the hypothesis.
  1. Formalize null hypothesis H_0
  2. Compute test statistic from data (e.g., number of heads)
  3. Compute *p* value
     - Conditional probability of test statistic taking on the computed or a more extreme value given H_0
  4. Reject H_0 if $p<\alpha$
     - Specify *p* as a measure of evidence against H_0
     - *p* does **not** provide a bound on the type I error, nor the posterior probability of H_0

6.2 t-Tests

Motivation: t-Tests

- Many random variables that are studied in neuroscience are (approximately) normally distributed



$$N\left(x \middle| \mu, \sigma^2\right) = \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- t-tests are appropriate for testing hypotheses about the means of normal distributions if *means and variances* have to be estimated from the data

Single-Sample t-Test

- In a **single-sample t-test,** we try to reject the null hypothesis that the mean of the distribution from which our samples were drawn equals some prespecified number
- *(Hypothetical) example:*
  Do chess players have larger brains on average?
  - H_0: μ=1260 (from literature); H_A: μ>1260 (one-sided)
  - Measured brain volumes of n=6 male chess players:
    $$\mathrm{v_i} = \{1400, 1220, 1280, 1360, 1290, 1350\}$$
  - Can we reject H_0 in favor of H_A?

Single-Sample t-Test: Computing t

1. Compute deviation of sample mean from $H_{0:}$

$$\bar{x} = \frac{\sum_{i=1}^{n} v_i}{\mathrm{n}} = 1316.7;\ \bar{x} - \mu_0 = 56.7$$

2. Compute sample variance:

"degrees of freedom"

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (v_i - \bar{x})^2 = 4266.7; s = 65.3$$

3. Compute standard error of sample mean:

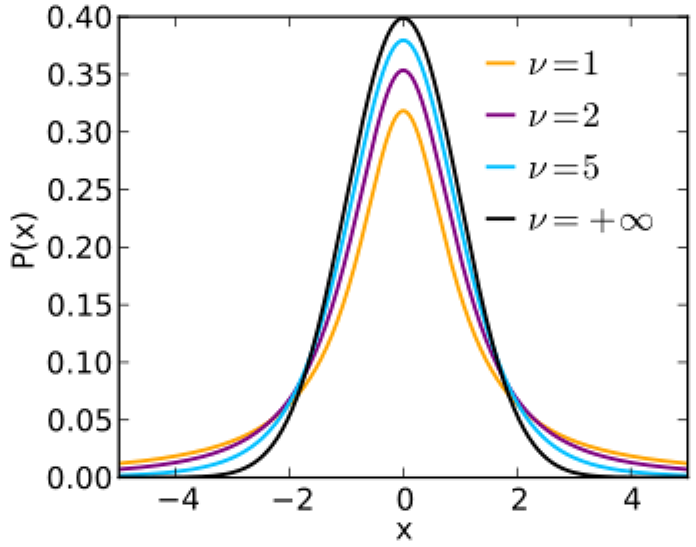
$$\frac{s}{\sqrt{n}} = 26.7$$

4. Compute *t* score, our test statistic:

$$t = \frac{\bar{x} - \mu_0}{s} \sqrt{n} = 2.13$$

Student's t-Distribution

- If we knew the true variance of $\bar{x} \sim N(\mu_0, \sigma_0^2)$, $(\bar{x} - \mu_0)/\sigma_0$ would follow $N(0,1)$
- Since we have to replace σ_0 with a sample estimate from *n* points (itself a random variable), t instead follows Student's t-Distribution with $\nu = n - 1$ degrees of freedom
- In practice, rely on software or lookup tables to convert t score and ν into a *p* value



Single-Sample t-Test: Final Result

- In our example, the one-sided *p* value for $t = 2.13$ and $\nu = 5$ is $p = 0.043$
  - Since $p < 0.05$, we reject H_0
  - We conclude that chess players have larger brains

- Alternatively, a lookup table would provide a “critical value” *q* against which *t* needs to be compared for a one-sided test, the given ν=5, and α=0.05
  - In our example, q=2.015 and $t>q$, so we reject H_0

From Single-Sample to Two-Sample

- **But wait:** We did not measure brain volume in the same way as the literature value against which we are comparing
  - Does our way of measuring provide comparable results?
  - To be safe, we measure the brains of six non-chess playing males using the same method and compare the two samples

$$v_i^{NC} = \{1190, 1210, 1310, 1370, 1250, 1230\}$$

  - Two-sample t-test: H_0: $\mu_C = \mu_{NC}$ H_A: $\mu_C > \mu_{NC}$

Two-Sample t-Test: Computing t

1. Compute difference of sample means:

$$\overline{x^{NC}} = 1260;\ \overline{x^{C}} - \overline{x^{NC}} = 56.7$$

2. Compute pooled sample variance (assuming same variances):

$$s_{NC}^2 = 4600;\ s_{C,NC}^2 = \frac{(n_C - 1)s_C^2 + (n_{NC} - 1)s_{NC}^2}{n_C + n_{NC} - 2} = 4433.3;\ s_{C,NC} = 66.6$$

3. Compute standard error of *difference of* sample means:

$$s_{C,NC}\sqrt{\frac{n_C + n_{NC}}{n_C \cdot n_{NC}}} = 38.4$$

4. Compute t score, our test statistic:

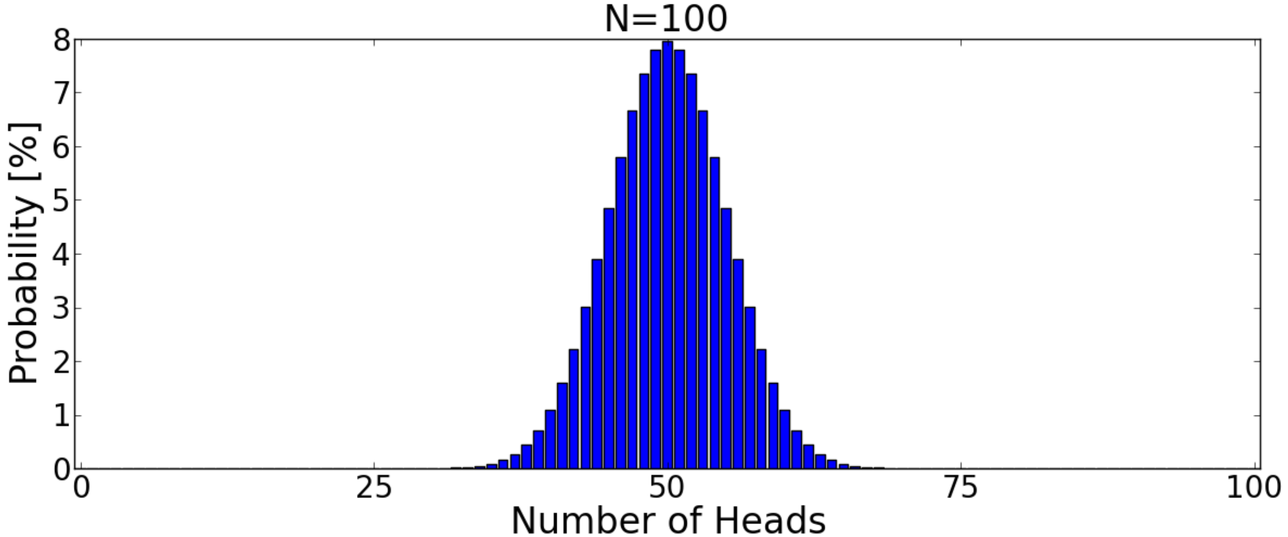
$$t = \frac{\overline{x^{C}} - \overline{x^{NC}}}{s_{C,NC}}\sqrt{\frac{n_C \cdot n_{NC}}{n_C + n_{NC}}} = 1.47$$

Two-Sample t-Test: Final Result

- Again, t follows Student's t distribution
  - In this case, $\nu = n_C + n_{NC} - 2$
  - One-sided *p* for t=1.47 and ν=10 is *p*=0.086
  - Test fails to reject H_0! **Why?**
    - We are now considering the difference of sample means, a random variable with larger variance.

t-Tests for Non-Gaussian Distributions?

- In practice, t-tests are sometimes applied even though the populations are not Gaussian distributed
- *Justification:* According to the **central limit theorem,** the sample mean $\bar{x}$ is expected to be "more Gaussian" than the original distribution
  - As $n \to \infty$, the average of n independent and identically distributed random variables with mean μ and (finite) variance σ^2 converges to a Gaussian
  - **Rule of thumb:** Requires sample sizes > 30-50



Two-Sample t-Test: Welch's Modification

- If variances of the two populations cannot be assumed to be equal, **Welch's Modification** computes:

$$t = \frac{\overline{x^C} - \overline{x^{NC}}}{\sqrt{\frac{s_C^2}{n_C} + \frac{s_{NC}^2}{n_{NC}}}}$$

  – Approximately follows Student's distribution with

$$\nu = \left(\frac{s_C^2}{n_C} + \frac{s_{NC}^2}{n_{NC}}\right)^2 \Big/ \left(\frac{(s_C^2/n_C)^2}{n_C - 1} + \frac{(s_{NC}^2/n_{NC})^2}{n_{NC} - 1}\right)$$

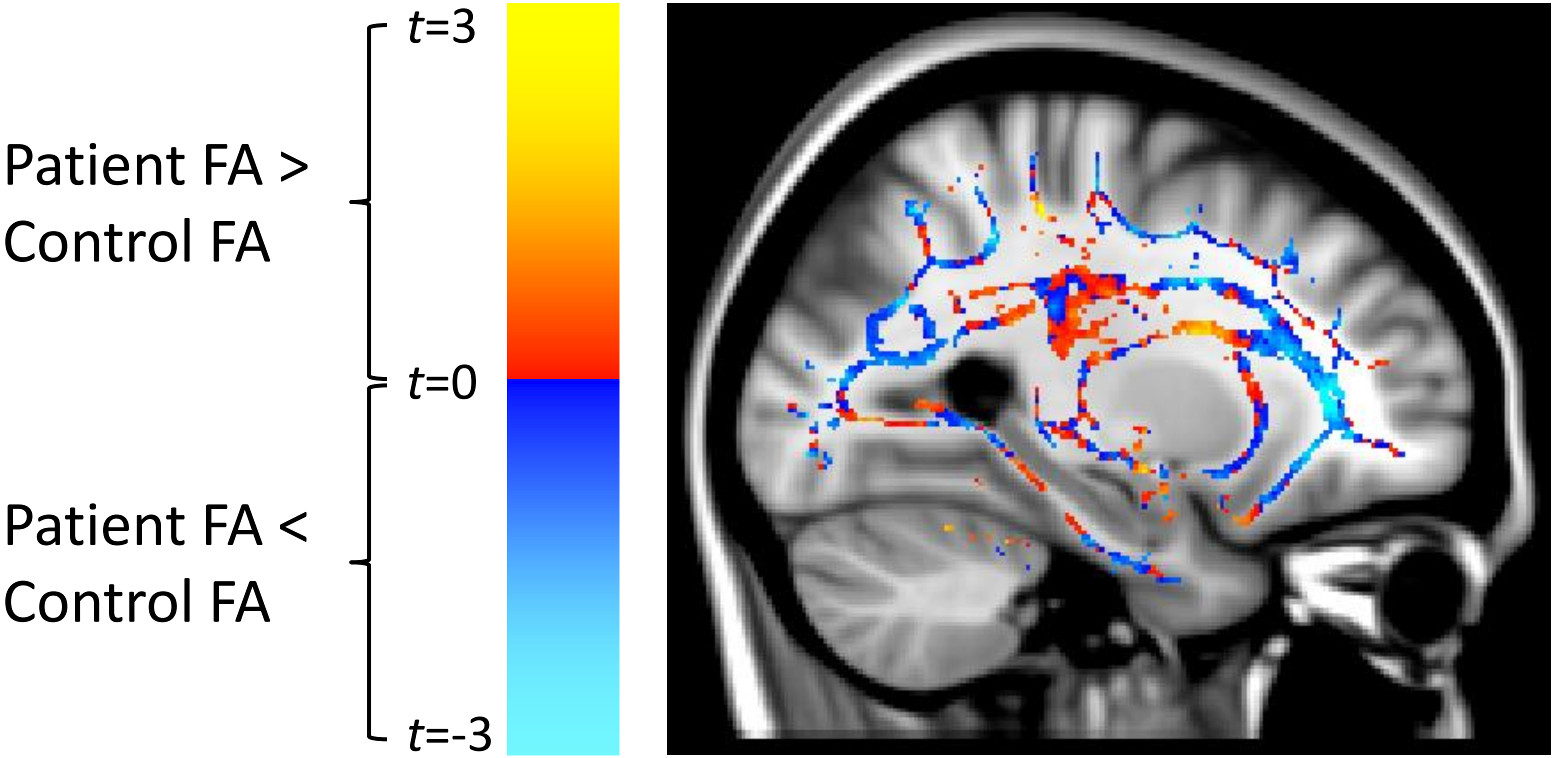
(rounded to the closest integer)

Paired t-Test

- Standard two-sample t-test assumes that **measurements are independent**. Examples in which this is violated include:
  - Repeated measurements of the same subject
  - Inclusion of siblings or even (identical) twins
- For pairwise associations, perform a test on the **difference between corresponding values** (paired test)
  - Example null hypothesis: Within a group of patients, there was, on average, no change in brain volume in a follow-up scan one year after treatment
  - Done by performing a single-sample t-test on the difference of corresponding values (e.g., after-before for each patient)

Example: Mass-Univariate Testing

- Performing one hypothesis test per voxel ("mass-univariate testing") is the most widely used way to draw conclusions in neuroimaging
  - Leads to statistical parametric mapping (SPM)
- Example:



Summary: t-Tests

- **t-Tests** are widely used to test for differences in means between Gaussian populations
  - Based on test statistics that follow Student's t Distribution
  - Applies when variance of sample means has to be estimated from the data
- **Variants:**
  - Single-sample: Compare sample mean to fixed value
  - Two-sample: Compare means of two groups
    - Welch's modification for unequal variances
  - Paired two-sample: Compare two groups in which measurements are associated pairwise

6.3 ANalysis Of VAriance (ANOVA)

Background: Family-Wise Errors

- **Quiz:** Suppose you have N=10 coins in your wallet and throw away all for which a statistical test with type I error level α=0.05 rejects the null hypothesis that it is legitimate. If none of the coins is counterfeit, what is the probability of still discarding at least one of them?

$$p = 1 - (1 - 0.05)^{10} = 0.40$$

- **Lesson:** If we perform multiple tests and would like to control the rate of performing a type I error in any of them (i.e., the **family-wise error**), we have to account for the number of comparisons!

Šidàk and Bonferroni Correction

- **Šidàk correction:** An obvious way to correct for multiple comparisons is to demand a stricter α_{IND}
  - Assuming independent tests, we obtain:

$$1-(1-\alpha_{\mathrm{IND}})^{N}=\alpha_{\mathrm{FW}} \text{ iff } \alpha_{\mathrm{IND}}=1-\sqrt[N]{(1-\alpha_{\mathrm{FW}})}$$

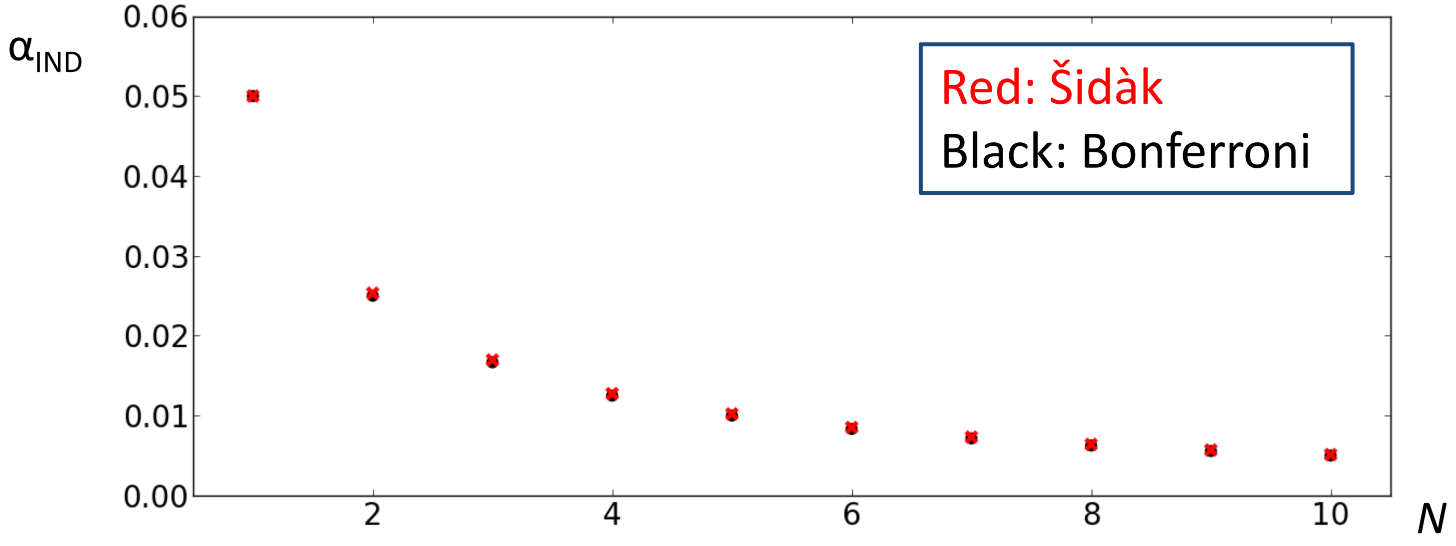
- **Bonferroni correction:** A computationally more convenient, conservative, and quite close approximation is given by:

$$\alpha_{\mathrm{IND}}=\alpha_{\mathrm{FW}}/N$$

Bonferroni Correction: Proof

- Consider N hypotheses $H_1,\dots,H_N$ with associated p values $p_1,\dots,p_N$. Let T be the set of N_T true hypotheses. Let $\alpha_{IND}:=\alpha_{FW}/N$. Then:

$$FWE = P\left(\bigcup_{i\in T}\left(p_i \le \frac{\alpha_{FW}}{N}\right)\right) \le \sum_{i\in T} P\left(p_i \le \frac{\alpha_{FW}}{N}\right) = N_T \frac{\alpha_{FW}}{N} \le \alpha_{FW}$$



Motivation: ANOVA

- Suppose you have a **factor with more than two levels** and would like to test for significant differences between their means
  - **Levels** could be different tasks or groups of people
  - **Potential approach:** Perform two-sample t-tests for each pair of levels
  - **Drawback:** Correcting for multiple comparisons quickly increases type II errors
- **Analysis of Variance** (ANOVA) performs a single test of the null hypothesis that means for all levels are equal. If it is rejected, we can investigate differences more closely.

ANOVA: Generative Interpretation

- y_{ij} (jth measurement in group i) is produced by adding the overall ("grand") mean μ, the effect a_i of belonging to group i, and random noise ε_{ij}:

$$y_{ij} = \underbrace{\mu + a_i}_{= \mu_i} + \epsilon_{ij}$$

$$\text{with } \sum_j \epsilon_{ij} = 0 \ \forall i \text{ and } \sum_i N_i a_i = 0$$

- In this interpretation, the null hypothesis is $a_i = 0 \ \forall i$
  - $a_i = \mu_i - \mu$, estimated as $\bar{x_i} - \bar{x}$
  - ANOVA can be interpreted as testing if a weighted sum of squared a_i (effects explained by the model) is significantly larger than the unexplained "noise" ($\varepsilon_{ij,}$ includes factors outside of our model)

Why Analysis of Variance?

- **Null hypothesis:** All groups are described by Gaussians with μ_1=μ_2=μ_3=μ and equal (but unknown) variance
- **Question:** Why analyze variance if we are interested in equality of group means?
- **Answer:** Partition overall variance σ^2 (of all data around μ) into between-group variance σ_{BG}^2 (between μ_i's) and within-group σ_{WG}^2 (around μ_i)
  - Observing $\sigma_{BG}^2 > \sigma_{WG}^2$ provides evidence against the null hypothesis. Therefore, the ratio $\sigma_{BG}^2/\sigma_{WG}^2$ will serve as the test statistic.

Partitioning the Sum of Squares

Interpretation in terms of "partitioning the sum of squares":

- Obviously:

$$y_{ij} - \mu = \left(y_{ij} - \mu_i\right) + \left(\mu_i - \mu\right)$$

Variation around grand mean

Variation around group mean

Variation of group means

- Interestingly, it is also true that:

$$\sum_{i,j} \left(y_{ij} - \mu\right)^2 = \sum_{i,j} \left(y_{ij} - \mu_i\right)^2 + \sum_{i} N_i(\mu_i - \mu)^2$$

Derivation: Partitioning the Sum of Squares

$$\sum_{i,j} (y_{ij} - \mu)^2 = \sum_{i,j} \left[(y_{ij} - \mu_i) + (\mu_i - \mu) \right]^2$$

$$= \sum_{i,j} (y_{ij} - \mu_i)^2 + \sum_i N_i (\mu_i - \mu)^2 + 2 \sum_{i,j} (y_{ij} - \mu_i)(\mu_i - \mu)$$

- The cross-term vanishes:

$$\sum_i \sum_j (y_{ij} - \mu_i)(\mu_i - \mu) = \sum_i (\mu_i - \mu) \underbrace{\sum_j \overbrace{(y_{ij} - \mu_i)}^{= \epsilon_{ij}}}_{= 0 \ \forall i}$$

ANOVA: Example

Assume the following (hypothetical) MR image intensities observed in the visual cortex during different tasks:

Task	Measurements		
1: Rest	1	2	2
2: Checkerboard	5	6	5
3: Beep	2	1	

Null Hypothesis: Mean intensity does not depend on the task

Notation:

- M=3 groups with N_i measurements each, adding up to N=N_1+N_2+N_3 measurements overall

ANOVA: Computing F

1. Compute sample estimates of group means μ_i and grand mean μ:

$$\overline{x_1} = 1.67; \overline{x_2} = 5.33; \overline{x_3} = 1.5; \bar{x} = 3$$

2. Compute between-group sum of squares:

$$SS_{BG} = \sum_{i=1}^{M} N_i (\overline{x_i} - \bar{x})^2 = 26.17$$

   – Degrees of freedom: *M-1=2*

3. Compute within-group sum of squares:

$$SS_{WG} = \sum_{i=1}^{M} \sum_{j=1}^{N_i} \left(y_{ij} - \overline{x_i}\right)^2 = 1.83$$

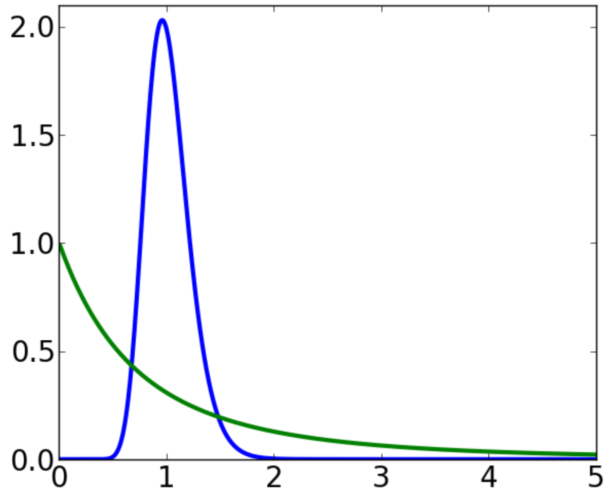
   – Degrees of freedom: *N-M=5*

4. Compute the ratio of both, divided by their respective degrees of freedom, as our test statistic:

$$F = \frac{SS_{BG}/(M-1)}{SS_{WG}/(N-M)} = 35.68$$

ANOVA: Testing F

- Under the null hypothesis, *F* follows the **F-distribution** with parameters d_1 and d_2
  - d_1 and d_2 are the degrees of freedom in the numerator / denominator, respectively
  - Again, we rely on software or tables to convert F, d_1, d_2 into a *p* value
  - Probability of $x \geq F$ provides evidence against H_0
  - In our example: *p*=1-CDF(35.68;2,5)=0.001
  - Since *p*<0.05, we reject the null hypothesis that mean intensity is the same during all three tasks



Blue: d1=d2=100
Green: d1=2, d2=5

ANOVA: Individual (“Post-hoc”) Testing

- If the F test failed to reject the null hypothesis that all means are equal, no further testing is done
- Otherwise, equality of individual means can be tested pairwise
  - Again need to correct for multiple comparisons!
  - Tukey’s HSD (“honest significant difference”) provides a conservative FWE-corrected critical value for the difference in means
    - *In our example:* At $\alpha=0.05$, HSD=1.74. Thus, “checkerboard” differs significantly from “rest” and “beep”, but the latter two do not differ significantly

ANOVA: The Two-Group Case

- With two groups and equal variances, the null hypothesis and assumptions in ANOVA are **equivalent** to a two-sample t-test
  - In this case, $F = t^2$
  - Two-sided two-sample *t*-test rejects null hypothesis μ_1=μ_2 iff *F*-test rejects it
  - *F*-tests do not give us a choice between one- and two-sided
    - Cannot test H_A: $\mu_1 > \mu_2$, since we lose the sign of *t*
  - *F*-test can be thought of as
    - *two-sided* in the sense that it assumes H_A: $\mu_1 \neq \mu_2$
    - *one-tailed* because only one end of the *F*-distribution is used

Two-Way ANOVA

- ANOVA can also deal with **more than one factor,** each of them at different levels (e.g., two groups of people each performing three different tasks)
  - Each combination (group x task) is called a “cell”
  - Data assumed to be Gaussian distributed (with the same variance) around cell means μ_{ij}
  - Two-way experiments typically strive for a balanced number of samples per cell (“full factorial design”)
- **Generative view:** y_{ijk} is kth measurement at level i of factor one and level j of factor two

$$y_{ijk} = \underbrace{\mu + a_i + b_j + (ab)_{ij}}_{=\mu_{ij}} + \epsilon_{ijk}$$

  - Allows us to test “main effects” a_i, b_j as well as “interactions” $(ab)_{ij}$

Summary: ANOVA

- **One-way Analysis of Variance (ANOVA)** used to test null hypotheses involving *multiple groups at the same time*
  - Partitions sum of squares into “between-group” (effect) and “within-group” (noise) parts
  - Test statistic follows *F* distribution
  - Generalizes two-sample *t* test
  - Additional post-hoc tests when null hypothesis was rejected
- **Two-way ANOVA** applies to factorial designs (such as multiple groups and tasks)
  - Can test for main effects and interactions

6.4 Family-Wise Error (FWE) Correction

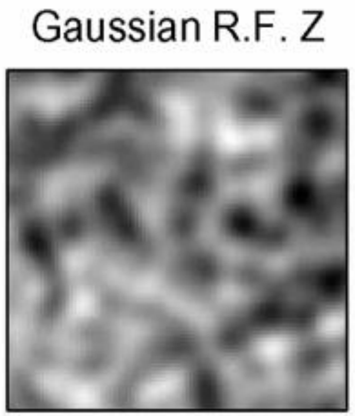
Family-Wise Errors in Statistical Parametric Maps

- **Statistical Parametric Maps** from neuroimaging can contain tens or hundreds of thousands of voxels
  - We are practically certain to encounter family-wise errors unless we correct for multiple comparisons
  - However, neighboring voxels are often correlated, making Šidàk and Bonferroni correction **far too conservative**
- **Ad-hoc approach:** Adapt threshold on test statistic until map "looks right" (e.g., t>3.1 or p<0.001)
  - Can attempt to empirically "calibrate" threshold based on a sufficient number of pilot runs in which the null hypothesis holds
  - Can be useful for quick visual screening, but non-trivial dependency on image resolution, details of processing pipeline etc.
  - *Not* a principled approach for reproducible and reliable conclusions

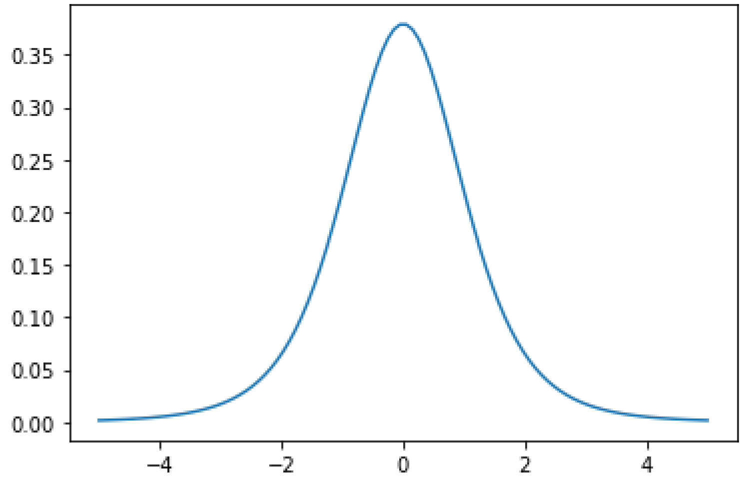
Introducing Random Field Theory

- **Random Field Theory** (RFT) studies the properties of a Gaussian random field $Z(\mathbf{x})$
  - Under the null hypothesis, Z has zero mean, unit variance
    - t scores are first transformed via (inverse) cumulative distribution functions

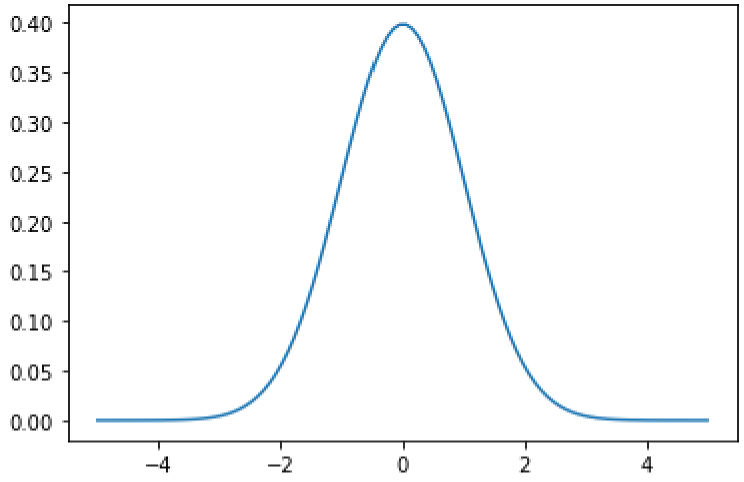
$$Z = \mathrm{CDF}_G^{-1}\left(\mathrm{CDF}_t(t)\right)$$



PDF Student's t ($\nu = 5$)



PDF Standard Gaussian

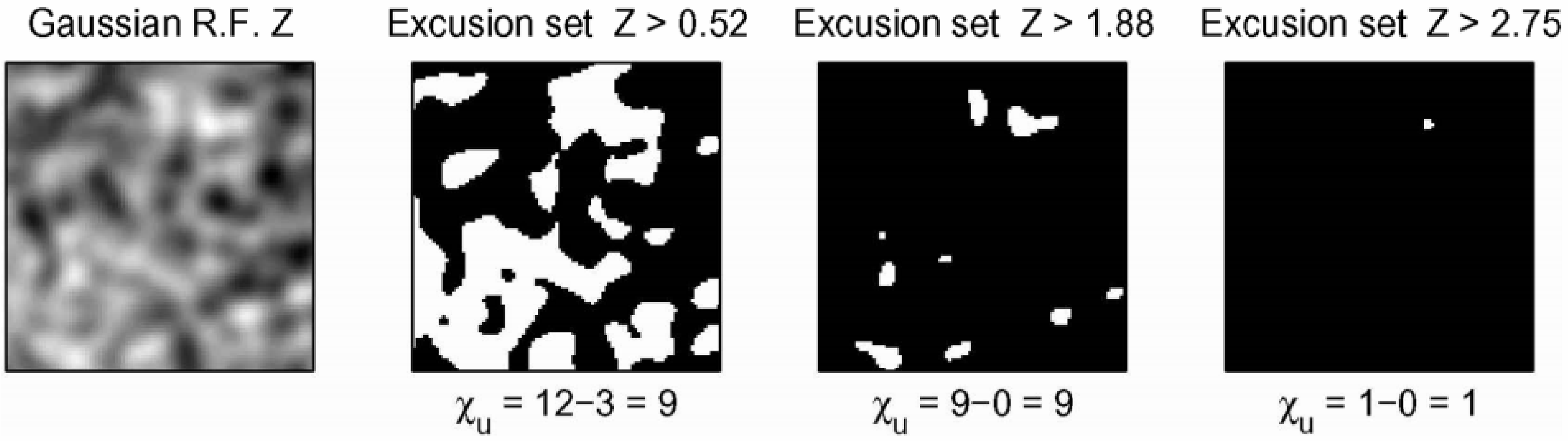


Images from [Nichols/Hayasaka 2003]

Voxel-Level FWE Correction with RFT

- Consider the **Euler characteristic** χ_u of the excursion set
$$\{\mathbf{x} \in \Omega \,|Z(\mathbf{x}) > u\}$$
  - In 2D, $\chi_u = \#\text{components} - \#\text{holes}$
  - Under the null hypothesis and for sufficiently large *u:*

$$p_{FWE}(u) = P\left(\bigcup_i Z_i \geq u\right) = P\left(\max_i Z_i \geq u\right) \approx P(\chi_u > 0) \approx E[\chi_u]$$

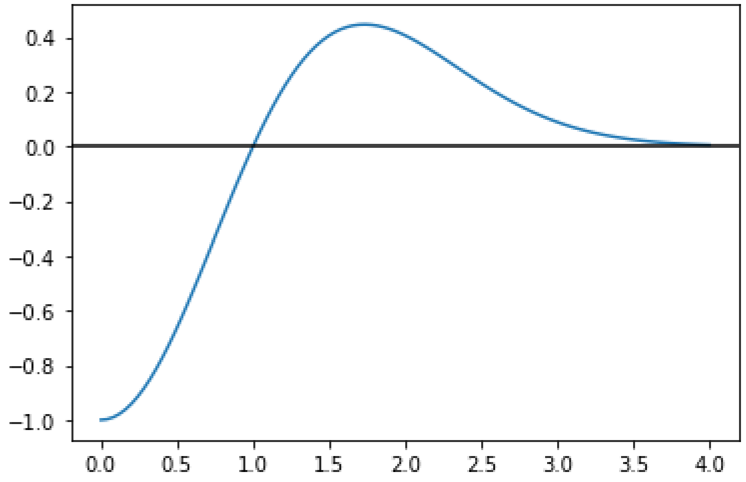


Images from [Nichols/Hayasaka 2003]

Voxel-Level RFT: Main Result

- **Theorem:**
  - *Assumption:* Z is defined on 3D volume V and is spatially smooth.
    - In practice, smoothness results from combined effects of intrinsic smoothness (due to physiology, PSF of acquisition) and explicit image smoothing
    - Λ denotes the variance-covariance matrix of its gradient ∇Z
    - Determinant $|\Lambda|$ indicates „roughness", smoother fields have lower values
  - For sufficiently large *u*,

$$p_{FWE}(u) \approx E[\chi_u] \approx \frac{V\sqrt{|\Lambda|}}{(2\pi)^2} \boxed{e^{-\frac{u^2}{2}}(u^2-1)}$$



Voxel-Level RFT: Resolution Elements

- Alternative statement of main theorem:

$$p_{FWE}(u) \approx R \frac{(4 \ln 2)^{\frac{3}{2}}}{(2\pi)^2} e^{-\frac{u^2}{2}} (u^2 - 1)$$

  - R is the number of resolution elements (RESEL)

$$R = \frac{V}{FWHM_x \times FWHM_y \times FWHM_z}$$

    where the full-width half-maximum (FWHM) refers to the hypothetical Gaussian kernel that *could have been used* to achieve observed smoothness from white noise
  - RESELS are introduced to aid intuition, voxel-level derivation does *not* assume squared exponential decay of correlation
  - Testing only in a subvolume *V* based on prior knowledge increases power

Cluster-level Testing

- **More powerful alternative to voxel-wise testing:** Consider size of connected regions ("clusters")
  - Most true effects extend over some spatial region, isolated voxels are likely to be false positives
  - Empirically, FWE-corrected cluster analysis is often more sensitive
- **Approach:**
  1. Form clusters by heuristically thresholding statistical maps at some level (e.g., $t>3$)
  2. Formal testing can be done **cluster-wise**
     - Size (number of voxels) or mass (sum of scores) serve as test statistics
     - **But:** How to determine null distributions?

Cluster-Based Testing with RFT

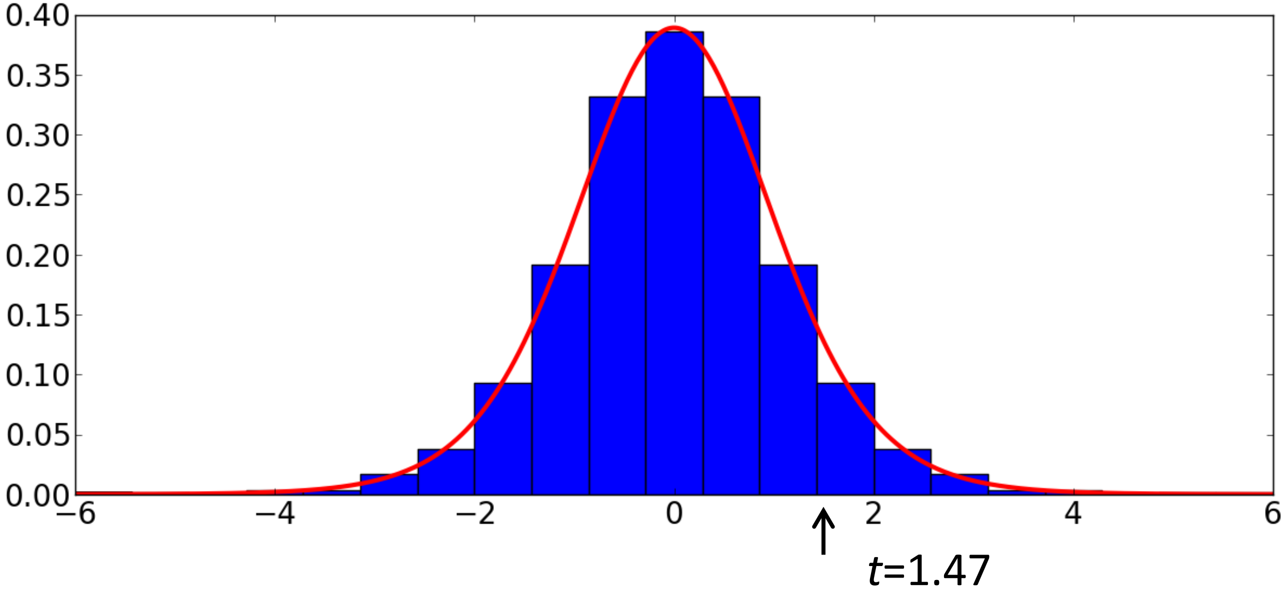
- **Random Field Theory** can also be used for cluster-based testing
  - Provides null distribution of cluster sizes
  - **But:** Introduces assumption of squared-exponential spatial autocorrelation
- [Eklund et al., PNAS 2016] demonstrate that **cluster-based RFT results are invalid**
  - Empirically found false positives in up to 70% of trials when null hypothesis was true, and nominal family-wise error rate set to 5%
    - In fMRI data from randomly selected resting subjects, spurious correlations were detected to an unrelated task
  - **Reason**: Heavy tails of real-world autocorrelation

Permutation-based Testing

- **Permutation methods** are widely used in neuroimaging to convert test statistics into *p* values
  - **Null hypothesis:** The assignment of individuals to groups (e.g., chess player vs. non-chess player) is unrelated to the value of the test statistic
  - Null distribution can be built by **permuting the group labels** w.r.t. the measurements and recomputing the test statistic for each configuration
  - *p* value given by location of test statistic for true label assignment w.r.t. null distribution

Illustration: Permutation-Based *p* Value

- Re-consider example of chess-player brain size:
  - $v_i^C = \{1400, 1220, 1280, 1360, 1290, 1350\}$
  - $v_i^{NC} = \{1190, 1210, 1310, 1370, 1250, 1230\}$
- The 12 measurements can be labelled in $\binom{12}{6} = 924$ ways, leading to the following distribution of *t* scores:



Red: Student's t-Distribution
p=0.086 (one-sided)

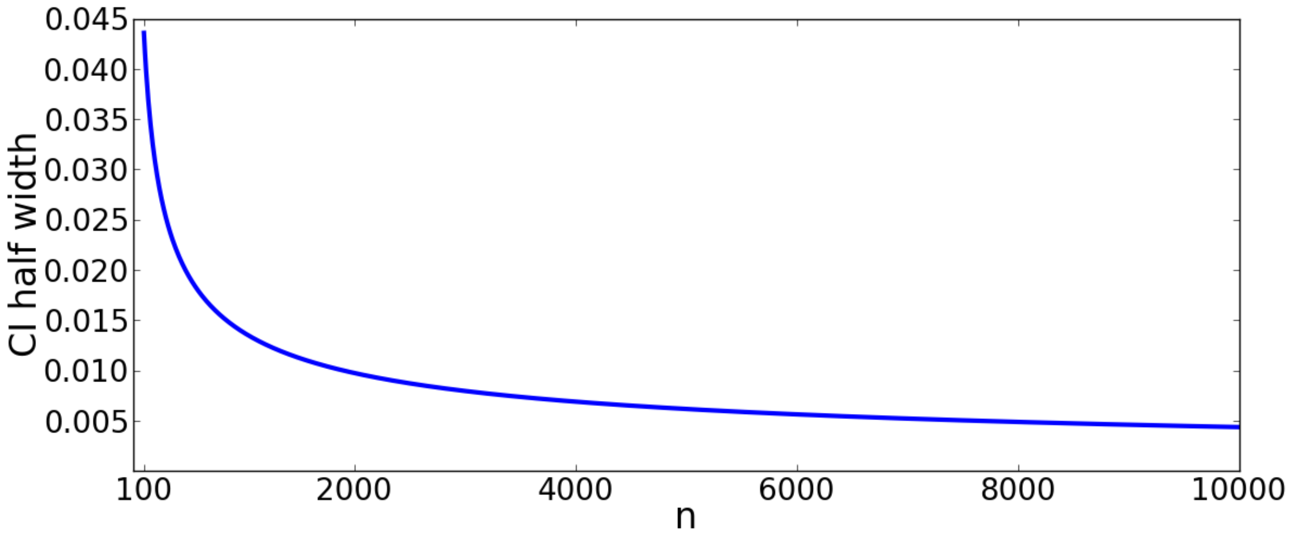
Blue: Permutation-based Distribution
p=0.091

- **Note:** Based on 924 possible permutations, no *p* values smaller than 1/924≈0.001 will be computed

Required Number of Permutations

- **Problem:** With more realistic sample sizes, total number of permutations becomes too large to enumerate
- **Solution**: Perform a random subsample
  - **Consequence:** Repeating the analysis of exactly the same data can lead to a different *p* value!
  - Randomly performing *n* out of a sufficiently large number of possible permutations produces the following 95% confidence interval (CI) of *p*:

$$p \pm 2\sqrt{p(1-p)/n}$$



Permutation-/Cluster-Based FWE Correction

- **For each permutation** of group labels:
  - Compute per-voxel test statistic (e.g., t-Test)
  - Threshold map at some level, compute clusters
  - Store the largest / heaviest cluster in the full brain
    - This is the relevant cluster to control FWE rate!

- **For each cluster** found using the true labels:
  - Compute per-cluster “FWE-corrected” *p* value by comparing size / mass to the null distribution

Reporting Cluster-Based Results

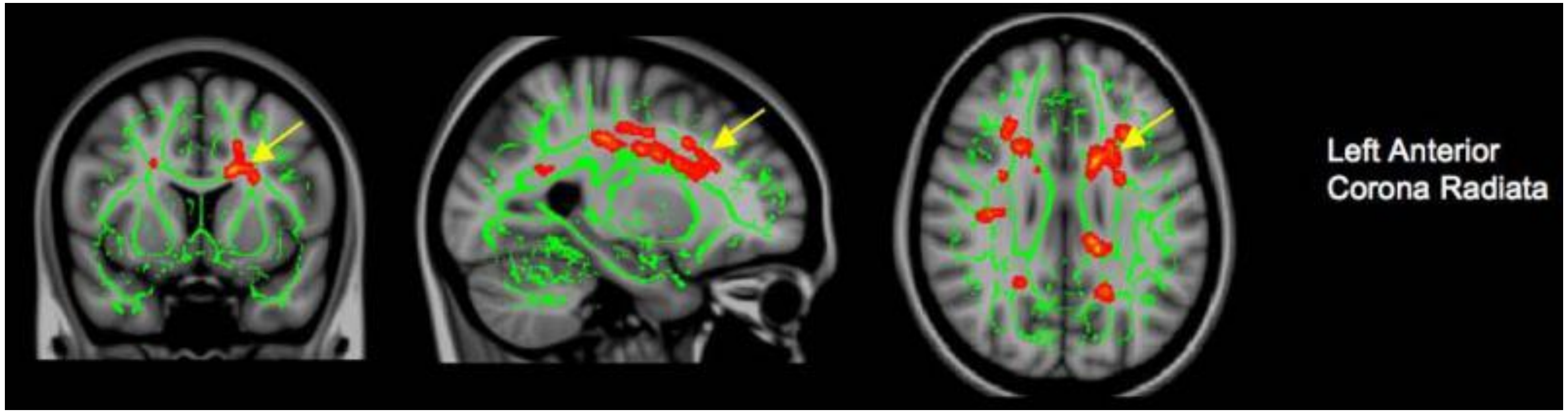
- *Example:* Table from [Jung et al. 2010]:

Table 3 DTI differences between NPSLE patients and controls

FA Controls > NPSLE

Voxels	p-value	Max-t	MNI X	MNI Y	MNI Z	Approximate white matter tract
266	0.001	3.135	30	17	18	Right superior longitudinal fasciculus
223	0.001	3.242	11	-34	25	Splenium of CC
172	0.003	3.408	-8	5	25	Body of CC
165	0.003	3.028	-26	28	12	Left anterior corona radiata
120	0.006	3.006	34	-41	28	Right superior longitudinal fasciculus

- Brain map from [Jung et al. 2010]:
  - Map *t* values, restricted to significant clusters



Threshold-Free Cluster Enhancement

- **Remaining drawbacks** of cluster-based testing:
  - No principled way to select cluster-forming threshold
  - Results may change considerably based on threshold
  - A few voxels might cause clusters to split or merge
  - Pre-smoothing commonly applied, but what bandwidth?
  - Center of gravity no longer an adequate description of localization if clusters are large
- Goals of **threshold-free cluster enhancement (TFCE)**:
  - Boost belief based on agreement in spatial neighborhoods
  - Avoid having to set an arbitrary threshold
  - Remains possible to perform testing per-voxel

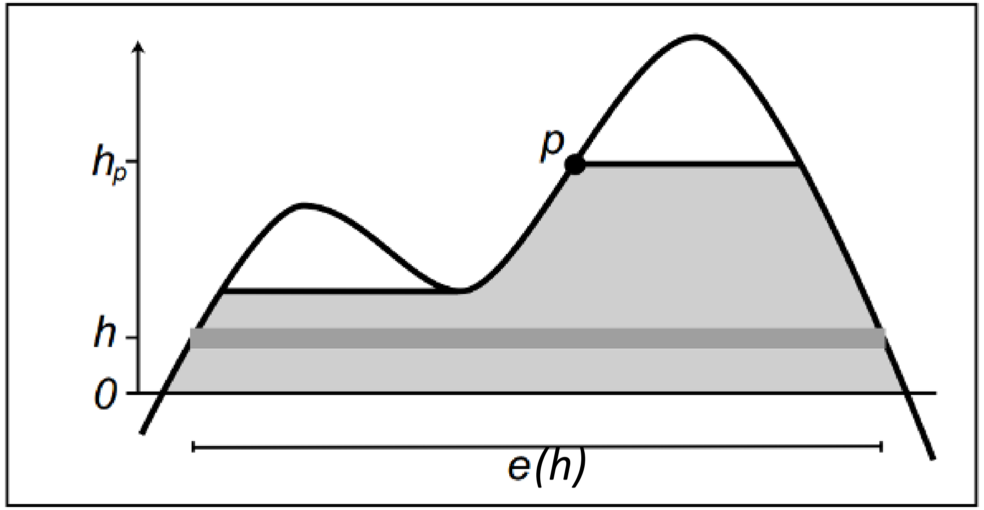
Basic Idea of TFCE

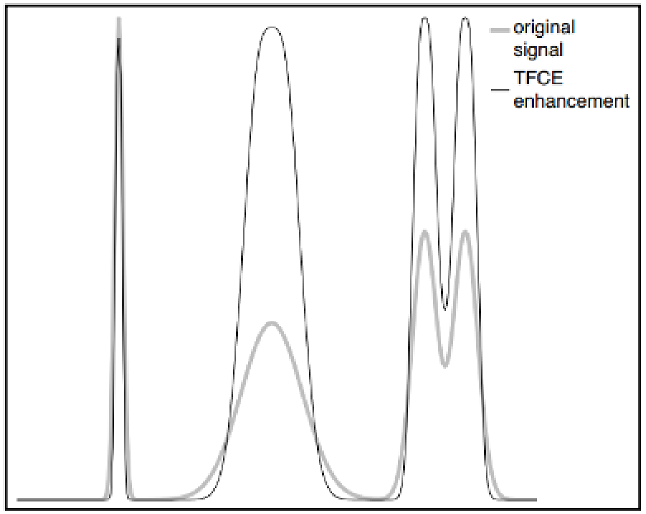
- **"Trick" in TFCE:** Perform image transformation that boosts values (e.g., *t* scores) that are surrounded by other large values

$$TFCE(p) = \int_{h=h_0}^{h_p} e(h)^E h^H dh$$

  – Recommended: E=0.5, H=2

Images from [Smith/Nichols 2009]





Properties of TFCE

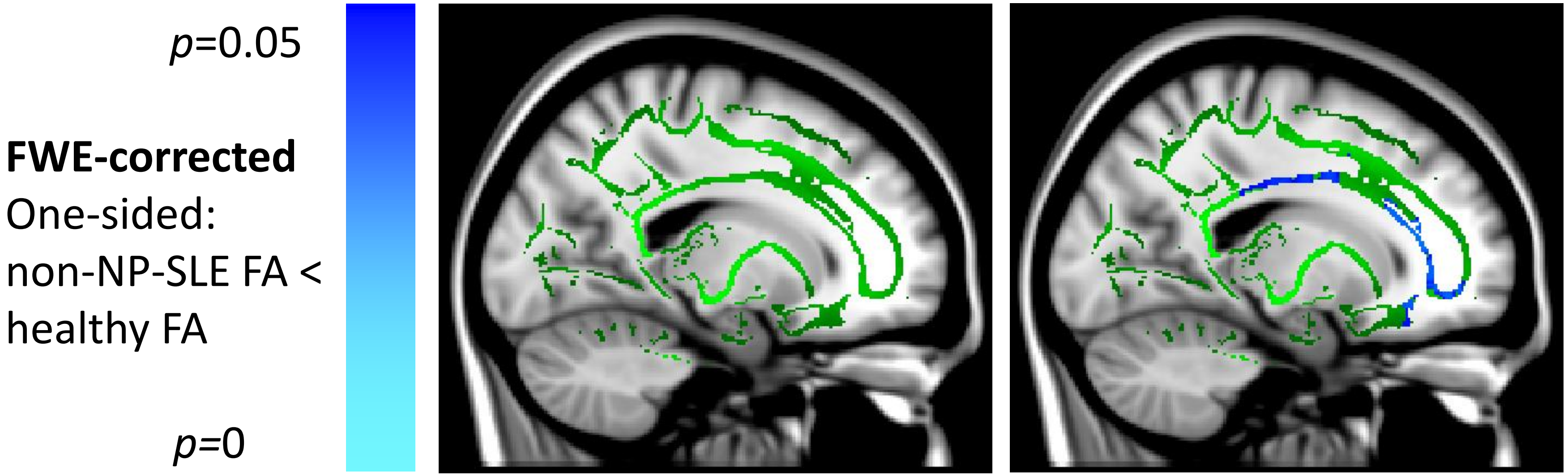
- TFCE only enhances **positive values,** sets negative ones to zero
  - Corresponds to a one-sided test
  - If desired, repeat test with negated test statistic (account for multiple comparisons!)
- Unlike Gaussian smoothing, TFCE **preserves locations of local maxima**
- **Permutation-based testing** using TFCE:
  - Compute per-voxel test statistic (e.g., t-Test)
  - Apply TFCE to the map of t values
  - Store the largest value in the full brain
  - Testing compares per-voxel TFCE value to the null distribution

Example: Cluster-based vs. TFCE

- *Example:* Systemic Lupus Erythematosus (SLE)
  - Autoimmune disease; would like to better understand difference between
    - NP-SLE: Neuropsychiatric variant, neural symptoms such as seizures or cognitive decline
    - non-NP-SLE: SLE without such symptoms
  - Using cluster-based analysis with cluster-forming threshold t>3, [Jung et al. 2010] conclude that NP-SLE (but not non-NP-SLE) leads to a significant difference in brain structure
    - Might lead us to believe that non-NP-SLE “does not affect the brain” (even though they neither tested nor claimed this!)

Cluster-based vs. TFCE in SLE

- Own results (based on different data):
  - Cluster-based analysis replicates Jung et al.
  - TFCE finds significant difference between healthy controls and non-NP-SLE!
    - Extent of affected regions much smaller
    - Suggests that even non-NP-SLE affects the brain, but small enough damage does not (yet) lead to neural symptoms



Cluster-based TFCE-based

[Schmidt-Wilcke et al., NeuroImage: Clinical 2014]

Summary: FWE Correction

- **Type-I errors accumulate** when performing multiple hypothesis tests
- **Random Field Theory** or **Permutation Tests** provide a way to correct for this while accounting for smoothness
  - Less conservative than Bonferroni correction
- Boosting confidence by using **information from neighboring voxels** can further increase sensitivity
  - Cluster-based
    - Cluster-forming threshold, permutation-based *per-cluster* FWE corrected *p* value
    - Random Field Theory for clusters should be avoided
  - Threshold-Free Cluster Enhancement (TFCE)
    - Nonlinear peak enhancement, permutation-based *per-voxel* FWE corrected *p* value

6.5 Voxel-Based Morphometry (VBM)

Goal of Voxel-Based Morphometry

- We just learned how to perform mass univariate tests, but how to derive images for which it makes sense to apply them?
- **Voxel-Based Morphometry** is one option
  - **Goal:** Would like to compare the size of specific brain areas between subjects
  - **But:** Segmenting specific regions of interest is non trivial and limits our analysis to those regions
  - **Idea:** Normalize subjects so that anatomical structures are aligned and compare regional gray matter volume *at each point of the brain*

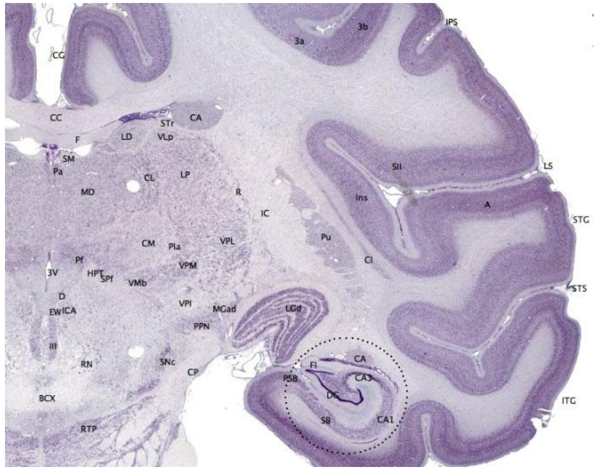
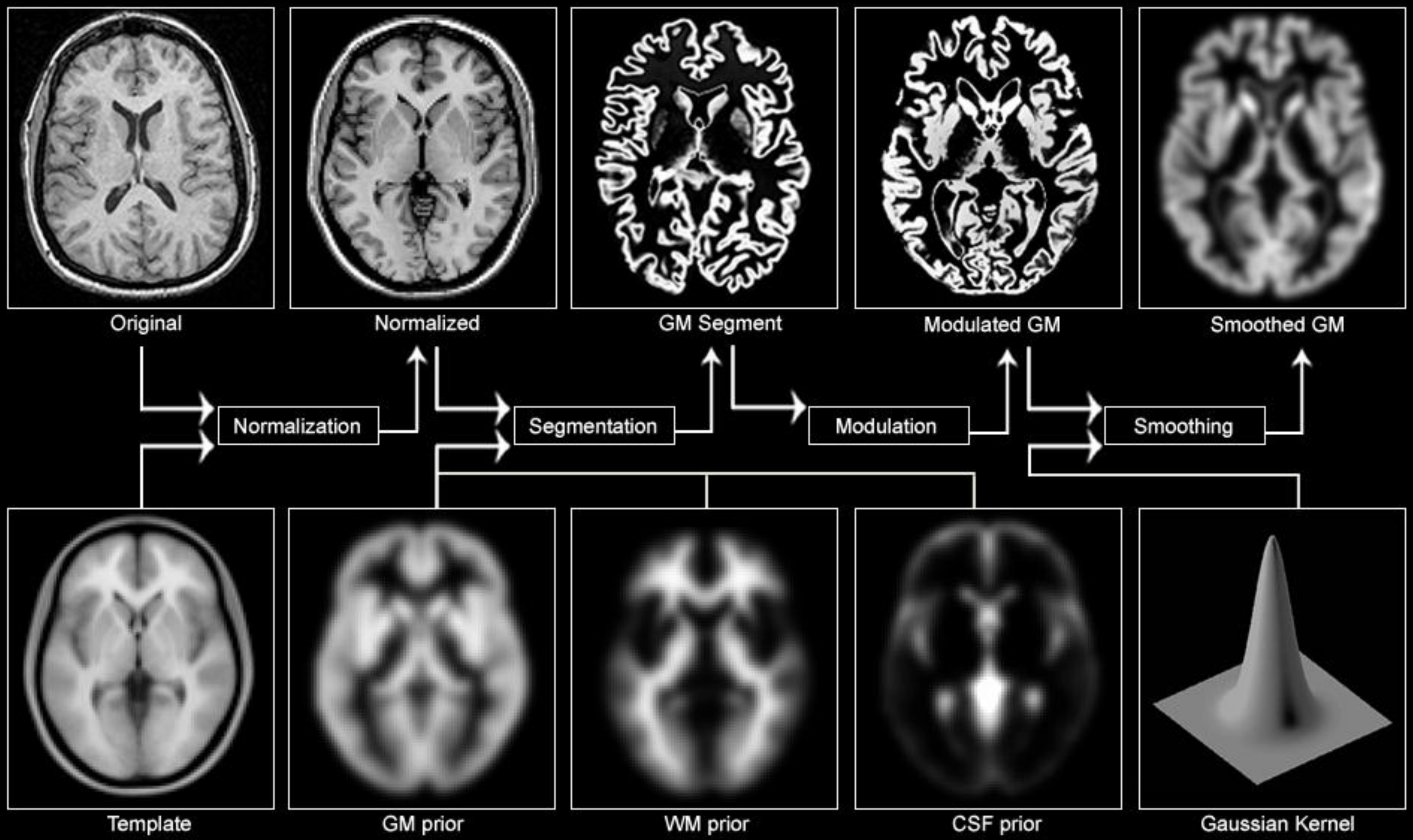


Image Source: brainmaps.org

Voxel-Based Morphometry
Pre-processing Overview



Slide from Nicola Hobbs & Marianne Novak

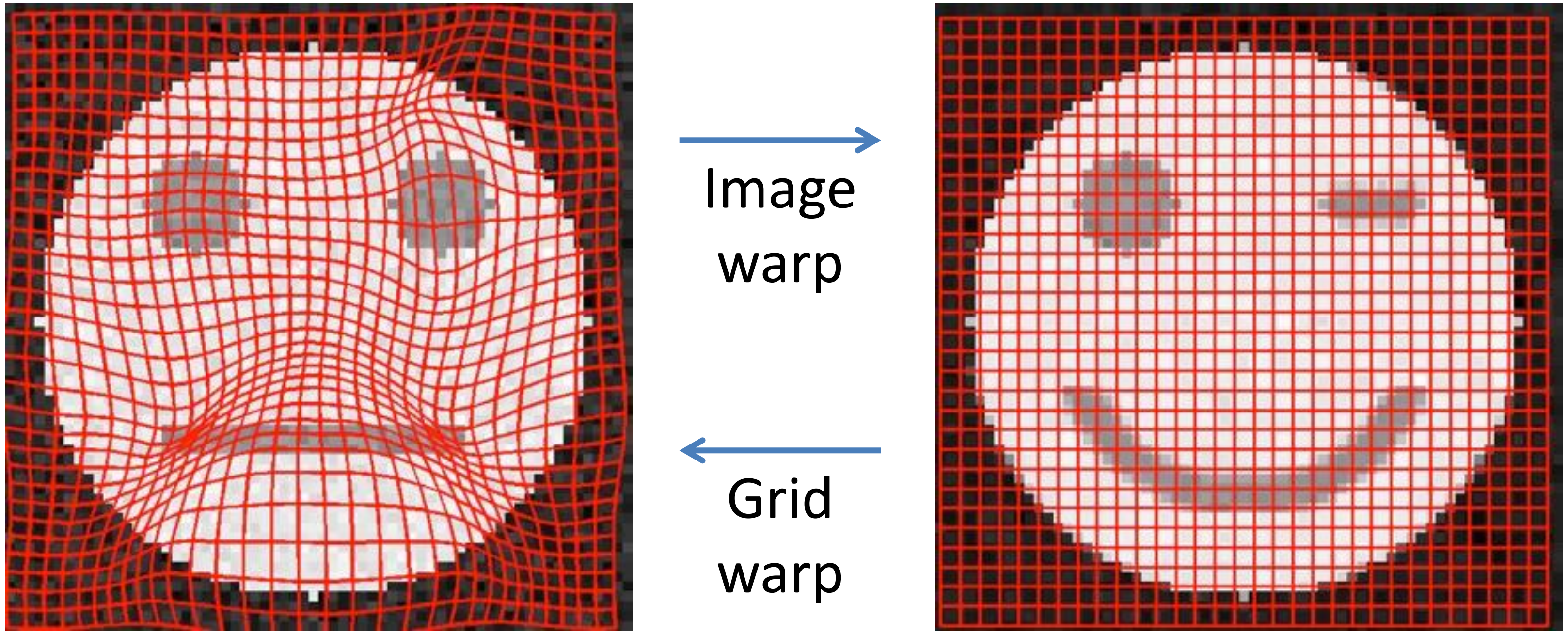
Pre-Processing in VBM

1. Normalization
   – Rigid, then affine, then non-linear registration
2. Segmentation
   – Tissue classification into GM / WM / CSF
   – MRF to deal with noise and bias fields

- **Optimized VBM:**
  – Performs automated brain extraction first
  – Builds a study-specific atlas
  – Achieves improved gray matter alignment by segmenting first, normalizing gray matter maps, applying transformation to original image, and segmenting again (now including prior maps)

Modulation

- Desired interpretation is in terms of **gray matter volume,** but nonlinear warping distorts volumes
- **Modulation** corrects for this by multiplying normalized gray matter map with the Jacobian determinant of the deformation field

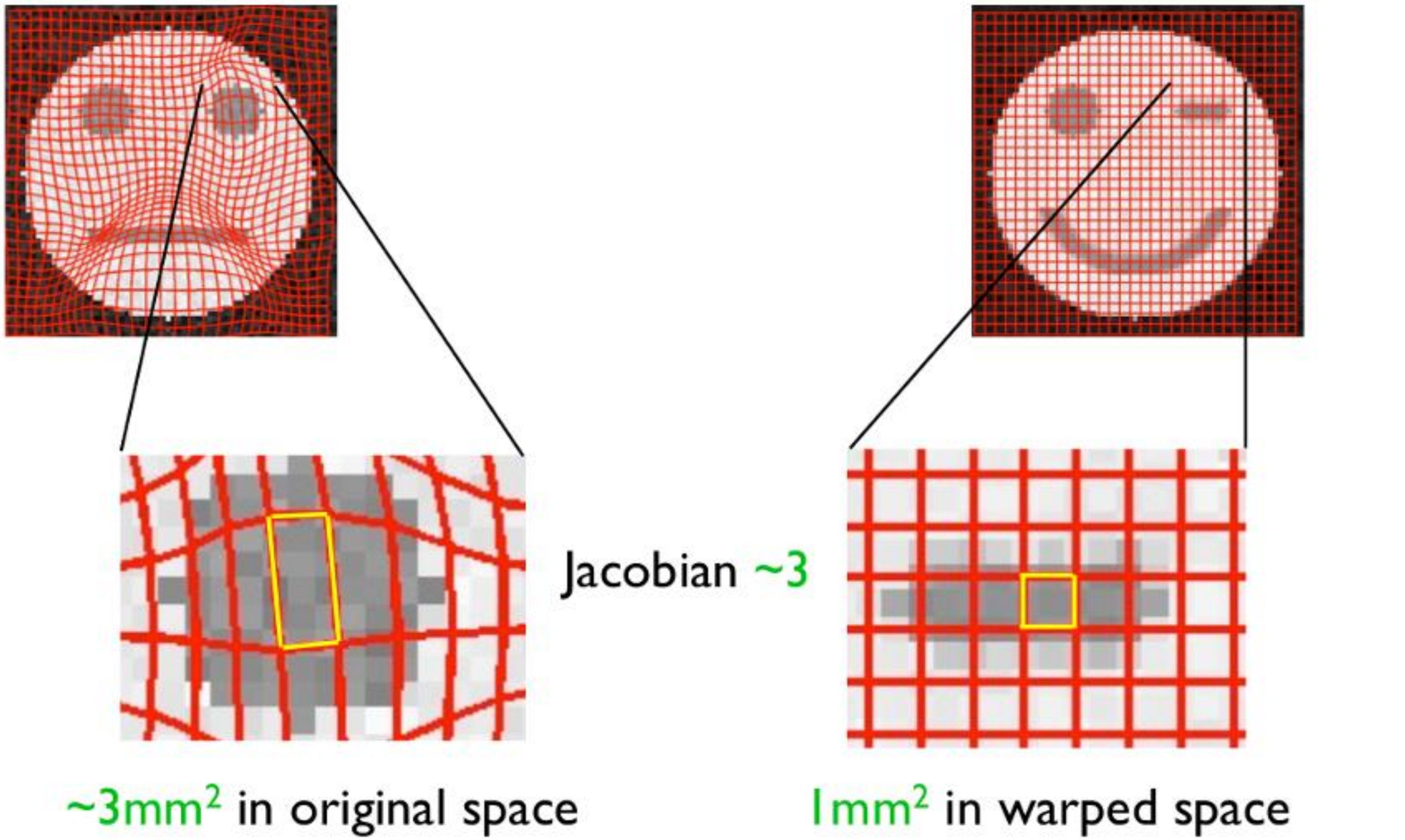


Individual Subject

Normal Space

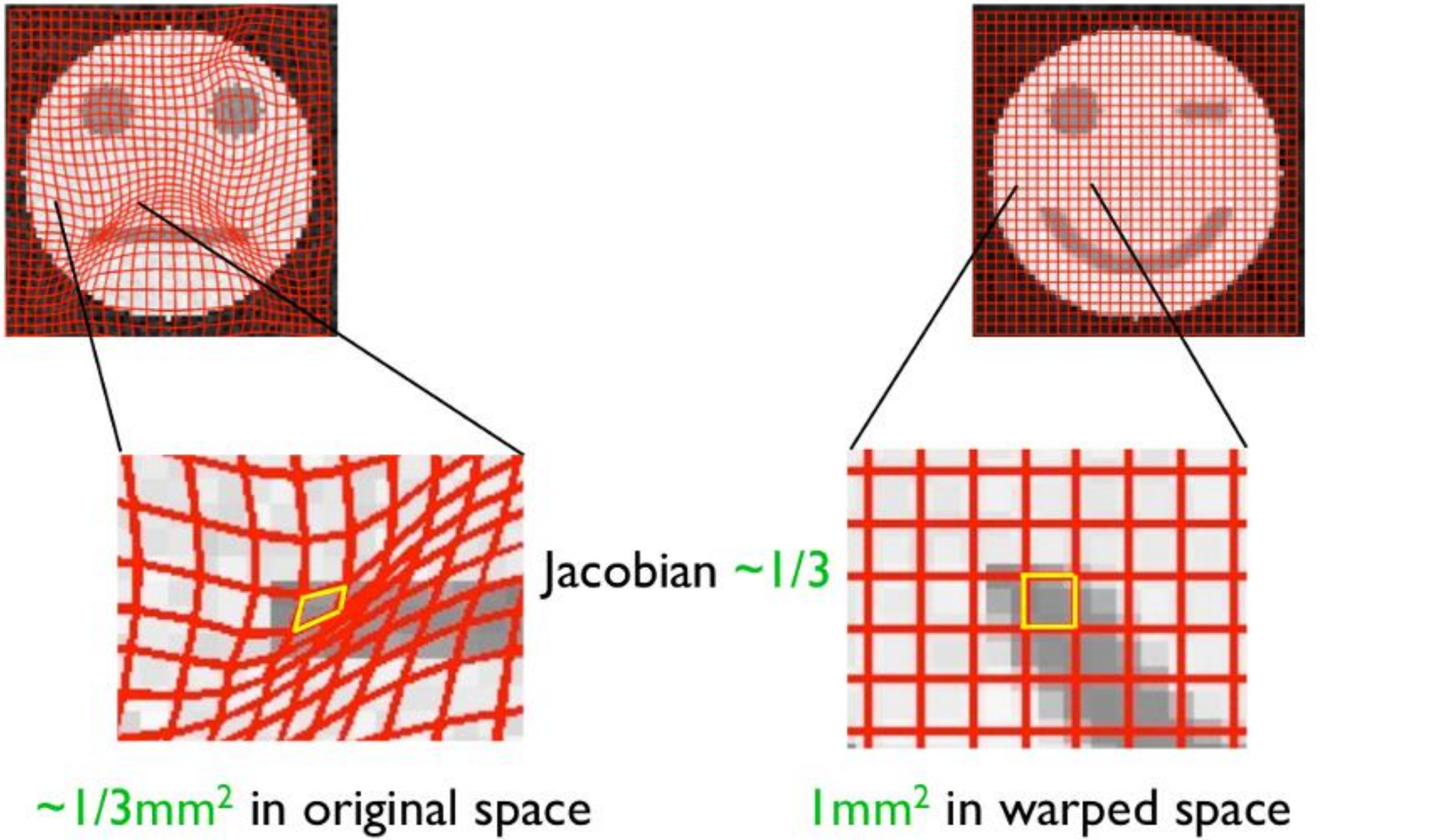
Images from Oxford brain fMRI lab

Illustration: Contraction



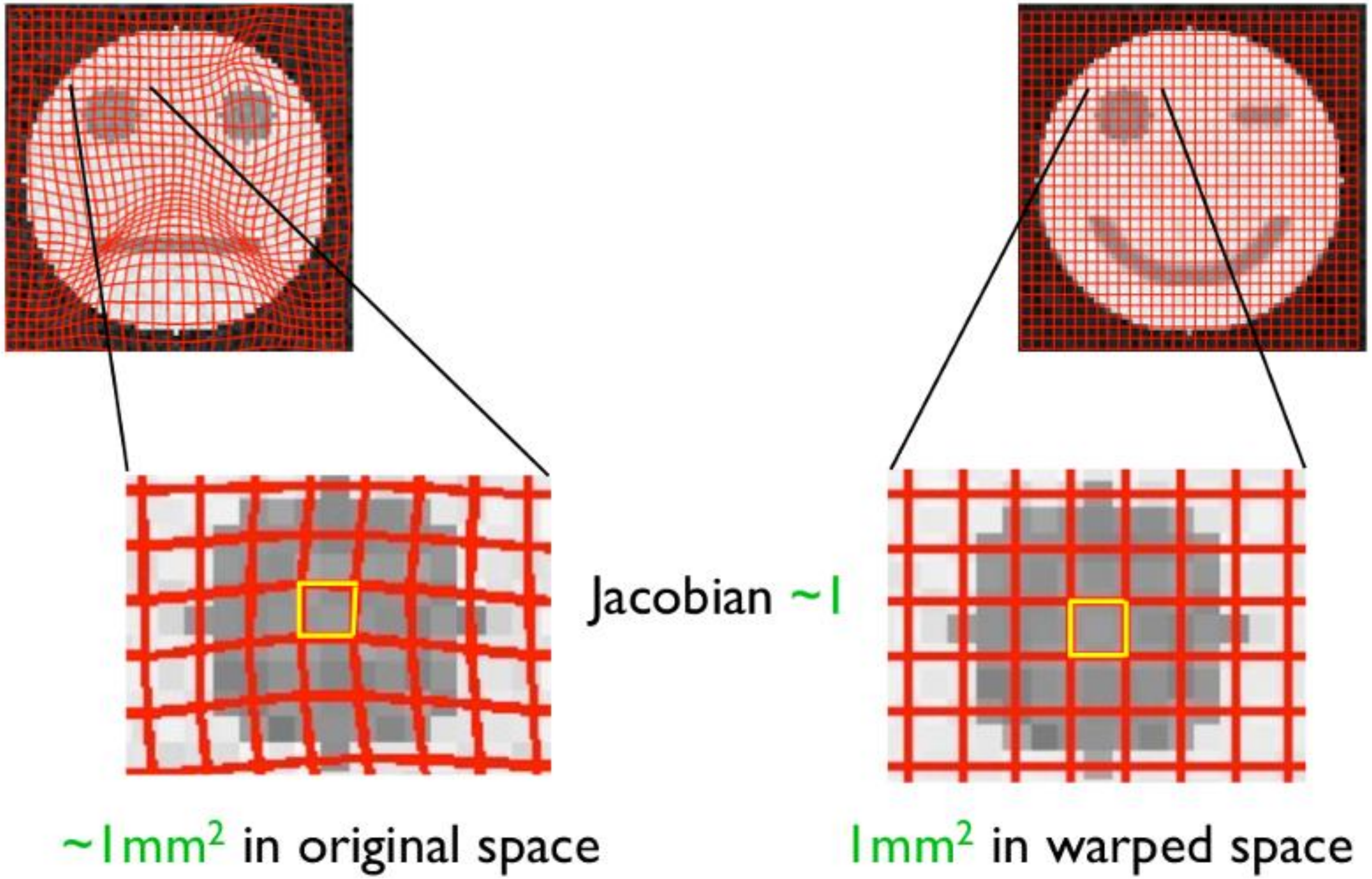
Slide from Oxford brain fMRI lab

Illustration: Expansion



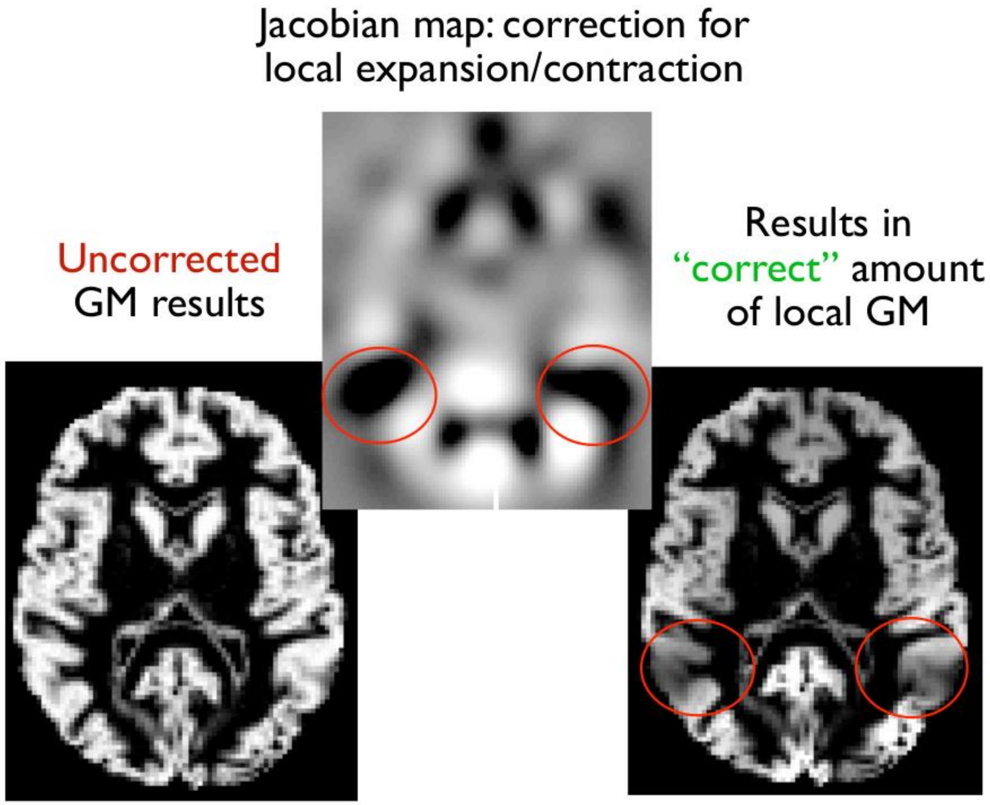
Slide from Oxford brain fMRI lab

Illustration: Volume Preservation



Slide from Oxford brain fMRI lab

Applying the Modulation

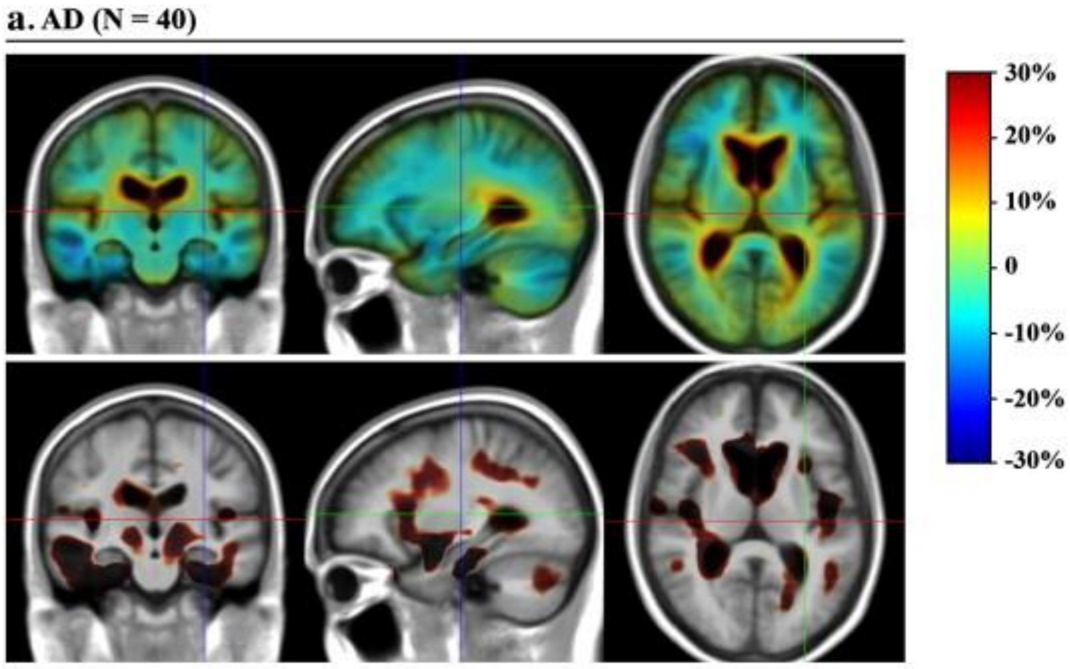


Images from Oxford brain fMRI lab

- Sometimes, modulation is omitted and results are interpreted as "gray matter density" (ratio of gray matter vs. white matter and CSF)

Tensor-Based Morphometry

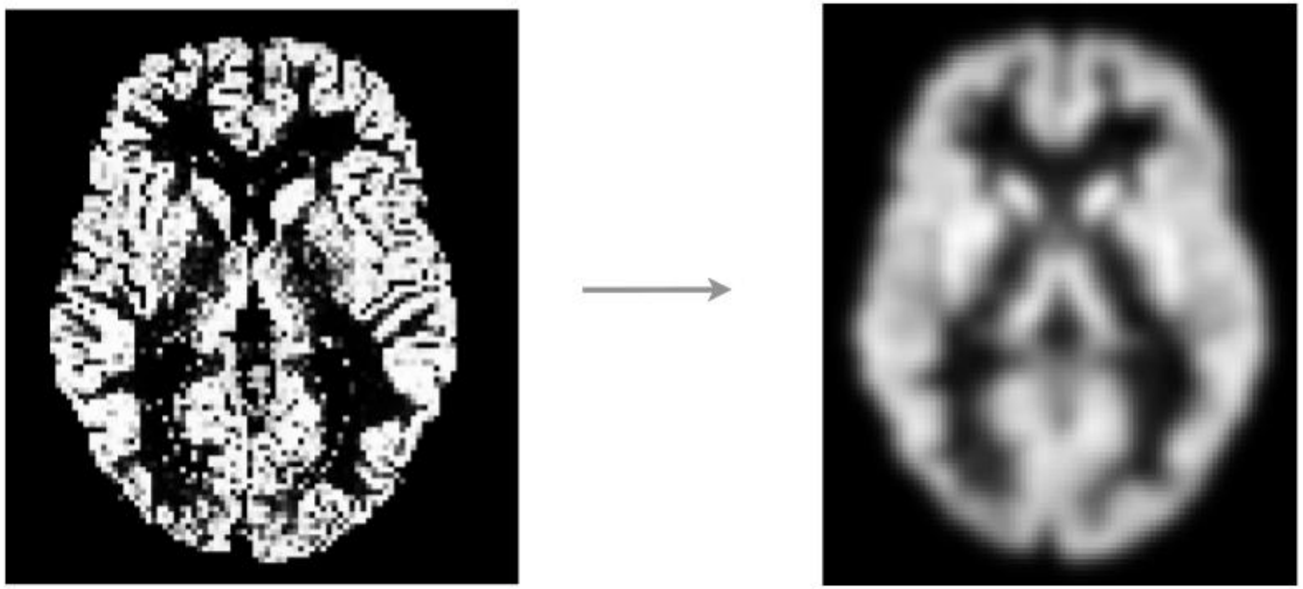
- **Tensor-Based Morphometry** (TBM) focuses on analyzing the deformation field
  - "Tensor" refers to Jacobian matrix
  - In the simplest case, based on Jacobian determinants
- *Caveat:* In simple VBM and in TBM, result will depend strongly on flexibility of normalization
  - VBM with modulation reduces this dependency



[Hua et al., NeuroImage 2008]

Smoothing

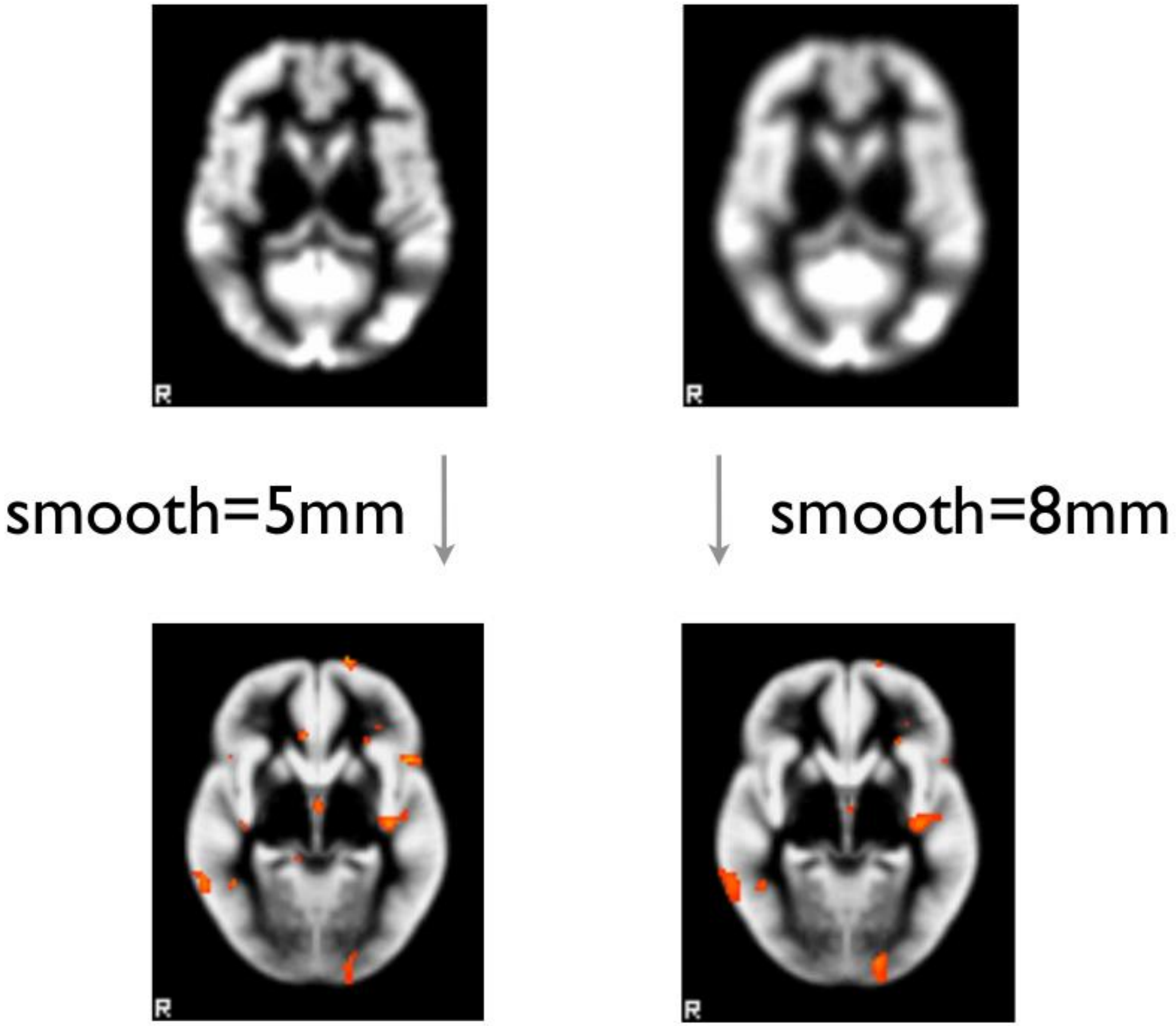
- Apply some amount of **Gaussian smoothing**
  - Interpretation as “regional” (rather than strictly local) gray matter volume / density
  - Compensates for slight inaccuracies in normalization
  - Makes the data more normally distributed (central limit theorem)



Images from Oxford brain fMRI lab

How Much Smoothing?

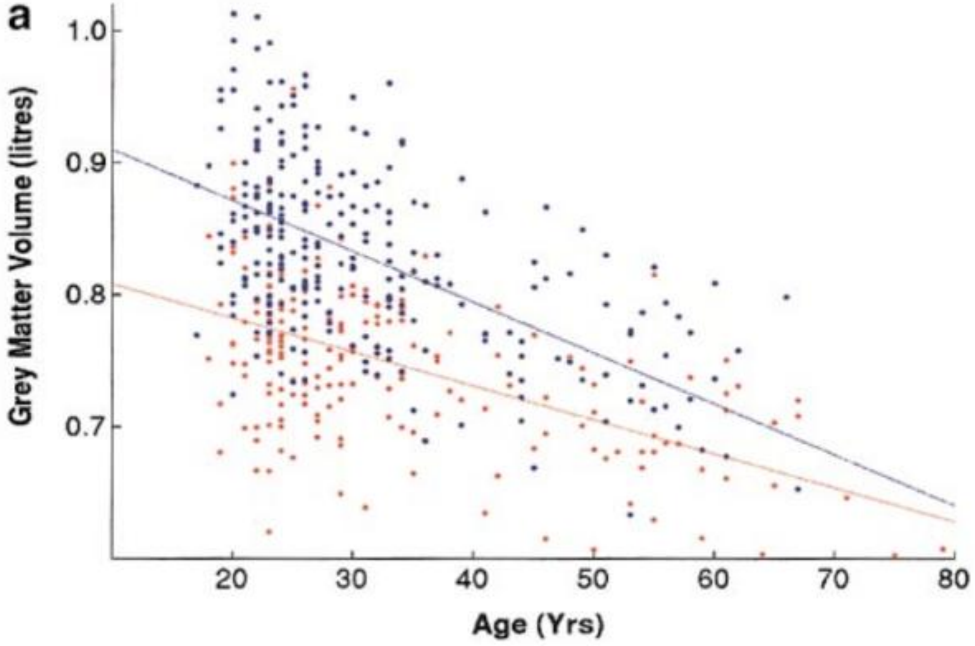
- Extent of smoothing will influence result; ideally, bandwidth should be tuned to expected size of effect

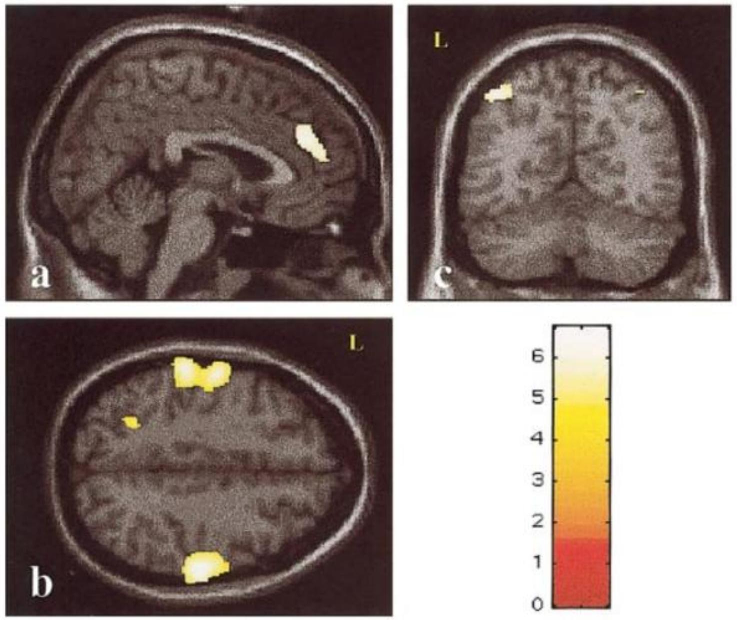


Images from Oxford brain fMRI lab

A VBM Aging Study

- [Good et al. 2001] used VBM to study the effect of age on global and local gray and white matter volumes in 465 healthy adults (200 female, 265 male)
  - Found global GM loss with age, accelerated in some regions, including pre-/postcentral gyrus

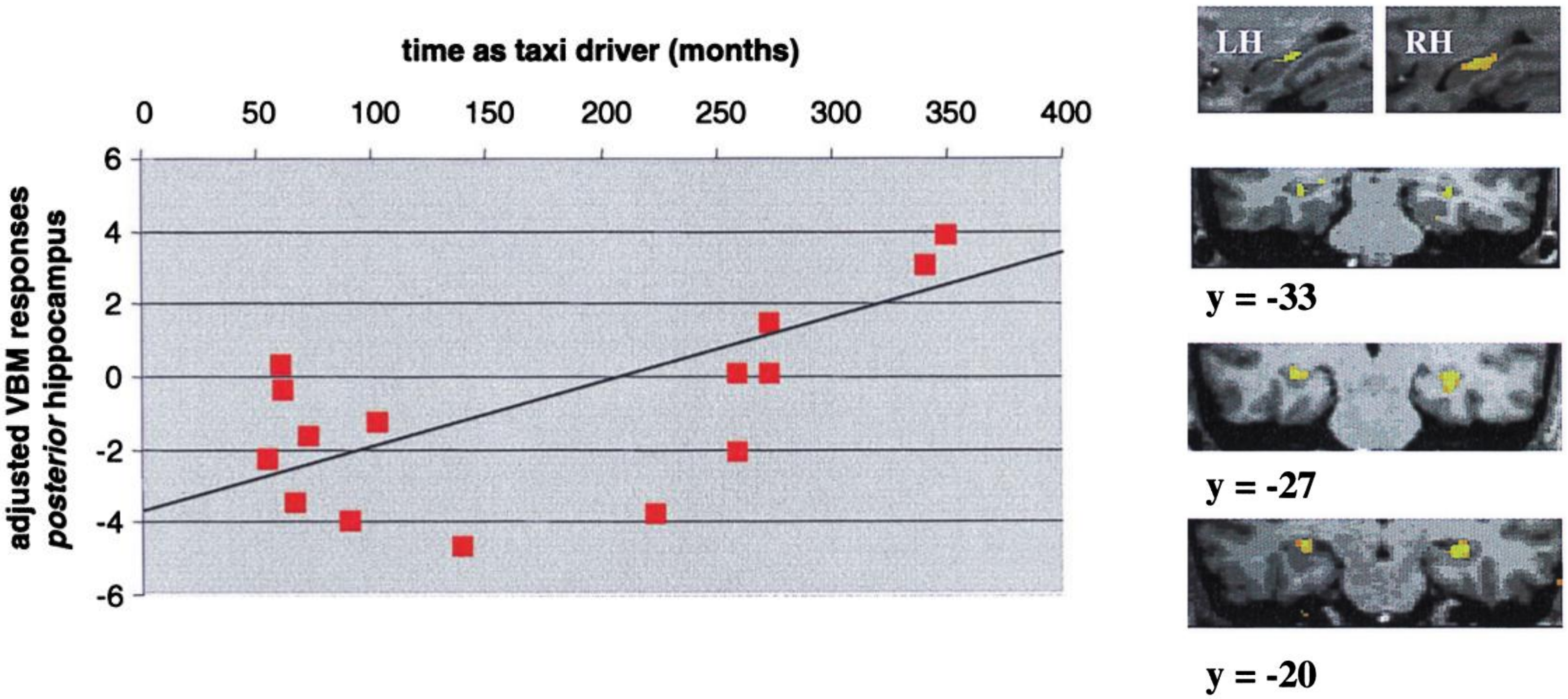




Accelerated loss of GM volume

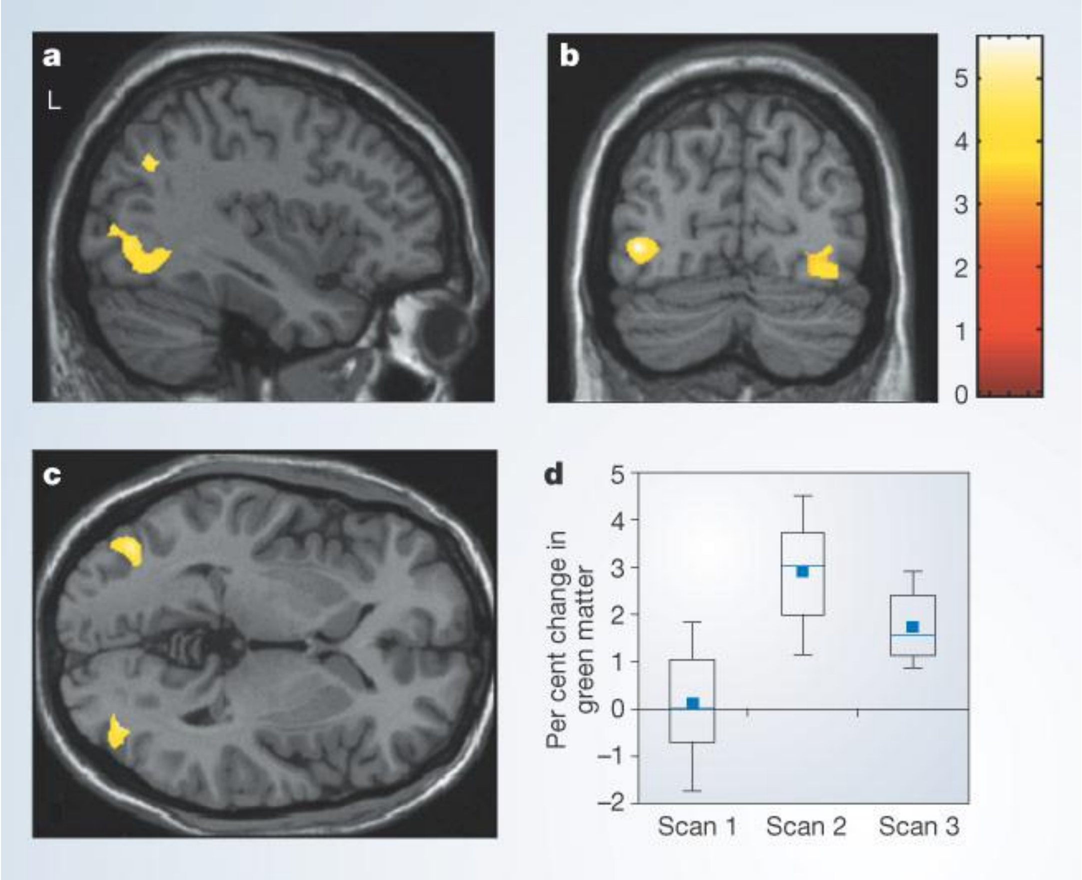
The VBM Taxi Driver Study

- [Maguire et al. 2000] used VBM to compare the brains of 16 taxi drivers to those of 50 gender- and age-matched non-taxi driving controls
  - Found enlarged posterior hippocampus in taxi drivers, correlated with time spent as a driver (age-corrected)



A Longitudinal VBM Juggling Study

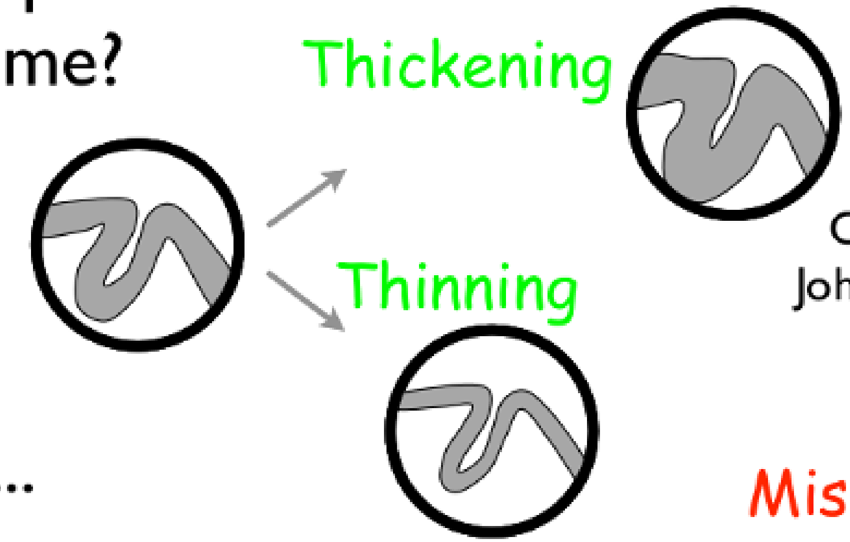
- [Draganski et al. 2004] used VBM to visualize transient structural changes in the brain associated with learning a new task (i.e., juggling)
  - Random assignment of subjects to groups, no significant differences initially
  - Changes localized in vision- and motor-related areas



Issues with Voxel-Based Morphometry

- Controversial approach - back to the issues:

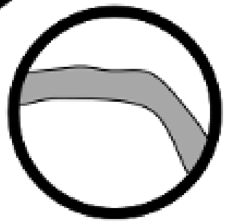
I) Interpretation of the results - real loss/increase of volume?



Or ...

- Difference in the contrast?

- Difference in gyrification pattern?



- Problem with registration?

Slide from Oxford brain fMRI lab

VBM: Summary

- **Voxel-Based Morphometry (VBM)** provides a tool to study regional changes in gray or white matter volume (or “density”) based on structural MRI scans. It involves:
  - Brain Extraction
  - Normalization
  - Segmentation
  - Modulation
  - Smoothing
  - Statistical Testing
- VBM is used widely and successfully
- Points of criticism include dependence on accuracy of registration and smoothing parameters

Further Reading Statistics in Neuroimaging

- Christian Heumann, Michael Schomaker, Shalabh: *Introduction to Statistics and Data Analysis*. Springer, 2016
- Nicole A. Lazar: *The Statistical Analysis of Functional MRI Data.* Springer, 2008
- Russell A. Poldrack, Jeanette A. Mumford, Thomas E. Nichols: *Handbook of Functional MRI Data Analysis.* Cambridge University Press, 2011

Further Reading FWE Correction

- T. E. Nichols, S. Hayasaka: *Controlling the familywise error rate in functional neuroimaging: A comparative review.* Stat. Methods in Medical Research 12(5):419-446, 2003
- S. J. Kiebel, J.-B. Poline, K. J. Friston, A. P. Holmes, K. J. Worsley: *Robust Smoothness Estimation in Statistical Parametric Maps Using Standardized Residuals from the General Linear Model.* NeuroImage 10:756-766, 1999
- S.M. Smith, T.E. Nichols: *Threshold-free cluster enhancement: addressing problems of smoothing, threshold dependence and localisation in cluster inference.* NeuroImage 44(1):83-98, 2009
- A. Eklund, T. E. Nichols, H. Knutsson: *Cluster failure: Why fMRI inferences for spatial extent have inflated false-positive rates.* PNAS 113(28):7900-7905, 2016

Further Reading VBM

- J. Ashburner, K.J. Friston: *Voxel-Based Morphometry – The Methods.* NeuroImage 11:805-821, 2000 [original VBM paper]
- C.D. Good et al.: *A Voxel-Based Morphometric Study of Ageing in 465 Normal Adult Human Brains.* NeuroImage 14:21-36, 2001 [optimized VBM paper]

Exam-like Questions

1. What is a Type I error in hypothesis testing? What is a Type II error? How do they relate to the frequently used level $\alpha = 0.05$?
2. Why is Bonferroni correction overly conservative in Statistical Parametric Mapping? Name a more sensitive, but still valid alternative.
3. Briefly explain the role of the modulation step in Voxel Based Morphometry.